

Prediction Market Accuracy: Crowd Wisdom or Informed Minority?

Roberto Gómez-Cram Yunhan Guo Theis Ingerslev Jensen Howard Kung*

This version: June 25, 2026

Abstract

Prediction markets produce remarkably accurate forecasts, but the source of this accuracy is poorly understood. Two explanations dominate: crowd wisdom and insider trading. Using the universe of Polymarket transactions, we show it is neither. Instead, accuracy comes from a minority of persistently skilled traders, around 3% of accounts. Unlike insiders, whose private edge is localized, these traders exhibit depth and breadth. They react to public news when it arrives, eliminate law-of-one-price violations, and trade against the crowd's behavioral mistakes. The crowd, in turn, generates most of the volume but little of the information, and its losses fund the minority's profits.

*Gómez-Cram, Guo, and Kung are at London Business School, Jensen is at Yale University; www.tijensen.com. We are grateful for helpful comments from Jiacui Li and Lasse Pedersen, as well as seminar and conference participants. Jensen gratefully acknowledges support from the Center for Big Data in Finance (grant no. DNRF167).

1 Introduction

Prediction markets have seen a sharp increase in volume, becoming an increasingly prominent part of financial markets. At their core, they price the probability of real-world events, forming a distinct class of financial instruments: event derivatives, binary contracts that pay \$1 if a specific event occurs and \$0 otherwise. Unlike traditional derivatives, which are based on broad financial variables such as asset prices or interest rates, event derivatives typically reference narrowly defined real-world outcomes, governed by precise rules and a fixed resolution date that pin down exactly what must occur and when. These prices are remarkably accurate in predicting a wide range of outcomes, from economic indicators and political elections to firm fundamentals, sports, and emerging domains such as technology and cryptocurrency. The source of this accuracy, however, remains poorly understood.

Two explanations dominate the debate. The first is the wisdom of crowds: prices aggregate the noisy signals of diverse participants, whose idiosyncratic errors cancel out as the crowd grows large. This is the dominant framing in both industry¹ and academia (Surowiecki, 2004; Wolfers and Zitzewitz, 2004; Manski, 2006). The second, voiced especially in the media,² is insider trading: with little regulatory oversight, pseudonymous accounts, and contracts on narrowly defined events, prediction markets are a natural venue for trading on private information. According to this view, accuracy reflects not the many but a few insiders who already know the outcome.

We show that neither explanation accounts for the pricing accuracy. Instead, it comes from a small minority of skilled traders. Using the universe of Polymarket transactions, we develop a measure of skill and find that only about 3% of accounts are skilled. Their trades predict future prices and final outcomes, moving prices toward the correct values. The crowd supplies most of the volume but little information, serving as the uninformed flow the skilled trade against. The skilled are also not merely insiders, since they continue to drive price

¹For example, Tarek Mansour (CEO, Kalshi) states that prediction markets “harness the power of the wisdom of the crowds” (Business Wire, 2026); Vlad Tenev (CEO, Robinhood) describes them as “the application of capitalism to the pursuit of truth,” where “market incentives and the wisdom of the crowds” sift through information (Crypto News Australia, 2025); and Shayne Coplan (CEO, Polymarket) argues that financial stakes aggregate information more effectively than experts do (CBS News, 2025).

²See recent media coverage of insider trading on prediction markets (Bruggeman, Katersky, and Simpson, 2026; CNBC, 2026; Wertheim, Chasan, Bast, and Kegu, 2026).

discovery even in markets not prone to insider trading. Across three proxies for inside traders, we find that insiders move prices sharply but in only a few markets, generating a tiny fraction of overall volume. The skilled, by contrast, trade across many markets and repeatedly within them, reacting quickly to news, enforcing the law of one price, and exploiting the crowd’s behavioral mistakes.

We reach these conclusions using transaction-level data from Polymarket, the largest prediction market platform by trading volume, whose monthly volume grew from \$3.3 million in December 2023 to nearly \$2 billion in December 2025. Polymarket settles trades on a public blockchain, which means that we have access to the universe of trades and accounts. In other words, we can see the exact contract bought or sold, the price paid, the size, and the timestamp down to the second, *and* we can link it to specific accounts. Such granularity is rare in finance, where account-level data are typically aggregated (e.g., the CFTC Commitments of Traders), reported at low frequency (e.g., quarterly 13F filings), or available only for a subset, such as retail investors from a specific brokerage firm (e.g., [Barber and Odean, 2000](#)).

Separating skill from luck is central to identifying who drives price discovery. A trader with a genuine information edge earns profits systematically, but others can earn large profits purely by luck, so realized profits alone cannot distinguish the two. We develop a sign-randomization test of skill tailored to the structure of prediction markets. Concretely, we take each account’s actual trades and randomly flip the direction (buy vs. sell) of each trade to generate a no-predictability null distribution. We randomize at the event level to account for correlations among contracts tied to the same event. An account’s profit is statistically significant if it falls in the tails of this null distribution. For example, a skilled trader is one whose realized profit is too high to be explained by random trading.

Using this test, we classify all accounts except market makers into four groups based on the significance and direction of their profits. *Skilled*, who earn significant profits, make up 3.2% of Polymarket accounts. *Lucky*, who earn insignificant profits, make up 28.9%. *Unlucky*, who incur insignificant losses, make up 61.7%. *Anti-skilled*, who incur significant losses, make up 6.2%. *Market makers* form a separate group, identified by their trading behavior rather than the skill test, and make up the remaining 0.1% of accounts. Together,

the lucky, the unlucky, and the anti-skilled form the crowd: the 96.7% of accounts whose trading shows no information edge.

We find an unusual degree of skill persistence. Splitting each trader's trading history into disjoint training and test sets, we find that 46% of traders classified as skilled in the training set are also classified as skilled in the test set. In a similar test of actively managed mutual funds, the corresponding number is 10%. For accounts classified as anti-skilled in the training set, 48% remain anti-skilled in the test set. The corresponding number for mutual funds is 6%. Luck is not persistent. For example, for accounts classified as lucky in the training set, 64% have losses in the test set. Realized profits alone, therefore, are a poor measure of skill. With millions of traders, some are bound to make large profits purely by chance. Our statistical measure of skill, by contrast, remains a powerful classifier out of sample.

Using the training-set classification, we test which groups' trades predict future prices and outcomes in the test set. We measure the directional trading activity of each group over short intervals by computing group-level order imbalance, the net buy volume of a group scaled by total trading volume. Skilled traders' order imbalance is a strong predictor of both next-period price changes and final market outcomes. The lucky, despite their positive profits, have no significant predictive power. The anti-skilled consistently trade against the eventual outcome, while market makers accommodate order flow in the short run but end up on the wrong side of informed trades at resolution, earning profits from liquidity provision rather than directional skill.

We next ask whether these predictive trades improve price accuracy. For each time bin, we measure the change in absolute pricing error and regress it on each group's share of trading volume. Skilled volume reduces pricing error throughout the market lifespan, and the effect grows substantially as resolution approaches. Other groups' trading is associated with prices moving further from the final outcome. Skilled traders, therefore, both anticipate price movements and also actively improve the informational content of prices. These results are robust to the choices behind our classification, including the benchmark that defines an uninformed trade and the criteria that identify market makers.

Could this skilled minority be insiders? We provide three pieces of evidence that suggest

otherwise. First, we use a large language model to flag the markets where insider trading is less plausible. These are markets where no individual can control the outcome or know it in advance, such as sports outcomes, cryptocurrency prices, and major elections. We find that roughly 90% of markets are non-insider-prone, and that skilled traders drive price discovery in these markets too. Their edge therefore cannot rest entirely on private information.

Second, the accounts that do look like insiders, flagged by proxies for suspicious timing, conviction, and track record, are few and localized, appearing in a small number of markets and accounting for less than 1% of volume. Third, we characterize how the skilled trade and find it inconsistent with insider behavior: they react quickly to public news, enforce the law of one price, and trade against the crowd’s behavioral mistakes, all strategies that rely on public information. Insiders are decisive when they act, but far too rare to move prices market-wide.

Our paper contributes to the growing literature on prediction markets. A large body of work establishes that prediction market prices are accurate forecasts of future events (Wolfers and Zitzewitz, 2004, 2006; Manski, 2006; Snowberg, Wolfers, and Zitzewitz, 2011), and more recent work uses these prices as real-time measures of market-implied expectations for macroeconomic indicators (Swanson, Wang, and Wu, 2025; Diercks, Katz, and Wright, 2026; Li and Xie, 2025) and firm-level earnings (Gomez Cram, Guo, Jensen, and Kung, 2026). Other papers using data from either Kalshi or Polymarket include Hartley (2025), Bürgi, Deng, and Whelan (2025), Sirolly, Ma, Kanoria, and Sethi (2025), Tsang and Yang (2026), Mitts and Ofir (2026), Glaze, López-Vilaró, and Stephan (2026), Della Vedova (2026), Yang (2026), and Fabi, Schönleber, Ruffo, and Marfè (2026).

The most closely related work studies trade-by-trade data, linked to accounts, on prediction markets. Ng, Peng, Tao, and Zhou (2025) examines price discovery across Polymarket, Kalshi, PredictIt, and Robinhood for contracts on the 2024 U.S. presidential election. Akey, Grégoire, Harvie, and Martineau (2026) documents a striking concentration of profits, with 1% of accounts capturing 76.5% of all gains. They also show that limit-order activity is a key determinant of profitability and argue that the 100 most profitable accounts are unlikely to be insiders. Reichenbach and Walther (2025) find that only one-third of accounts earn positive profits and present evidence that the distribution of profits across all accounts is

consistent with some accounts having skill.³

Our contribution is to ask who makes prediction markets accurate. We depart from existing work in two ways. First, we measure skill rather than realized profits: with hundreds of thousands of accounts, many large winners are merely lucky, and we show that realized profits are a poor proxy for skill. Second, we link skill directly to price discovery, asking which traders move prices toward the eventual outcome. The answer is a small minority of persistently skilled traders who trade on public information. This answer goes against conventional explanations for prediction market accuracy, which tend to emphasize crowd wisdom or insider trading.

We also contribute to the literature on retail trading and price informativeness (Barber, Odean, and Zhu, 2008; Kaniel, Saar, and Titman, 2008; Barrot, Kaniel, and Sraer, 2016; Hvidkjaer, 2008; Kelley and Tetlock, 2013; Boehmer, Jones, Zhang, and Zhang, 2021; Barber, Huang, Odean, and Schwarz, 2022; Welch, 2022; Eaton, Green, Roseman, and Wu, 2022). That literature typically relies on indirect measures of retail activity or on direct measures from a subset of participants. In prediction markets, we observe the universe of trades and accounts, so we can classify every trader directly by role and performance. We find that the vast majority do not contribute to price discovery, and that accuracy comes instead from a small minority of skilled traders. Our results also speak to work linking portfolio concentration to informed trading among individual investors (Ivković, Sialm, and Weisbenner, 2008) and mutual funds (Kacperczyk, Sialm, and Zheng, 2005). In prediction markets, by contrast, the most concentrated winners are usually lucky rather than skilled, a difference we attribute to the idiosyncratic, one-shot nature of events.

Our statistical measure of skill in prediction markets contributes to the literature on measuring skill in financial markets, beginning with Jensen (1968) and including Carhart (1997), Kosowski, Timmermann, Wermers, and White (2006), Fama and French (2010), Barras, Scaillet, and Wermers (2010), Berk and Van Binsbergen (2015), and Kacperczyk, Sialm, and Zheng (2008). These papers typically measure skill from fund-level returns,

³Bossaerts, Yadav, Bossaerts, Nash, Todd, Rudolf, Hutchins, Ponsonby, and Mattingly (2024) is another related paper in the sense that they study skill in prediction markets, but focus on the academic markets rather than Polymarket or Kalshi. They find evidence that traders who are sensitive to prices do better than random chance.

whereas we observe every individual trade. A unique aspect of our skill measure, therefore, is that the randomization is performed at the trade level while accounting for the rich correlation structure of trades across markets within the same event.

Finally, our work also connects to the theoretical literature on informed trading. [Grossman and Stiglitz \(1980\)](#) provides the canonical framework for price informativeness in the presence of costly information and noise, while [Kyle \(1985\)](#) and [Glosten and Milgrom \(1985\)](#) establish the core market microstructure foundations for how informed traders interact with noise traders and market makers. We build on this literature to formally distinguish the wisdom of crowds from informed trading. Our results challenge the conventional view that prediction-market accuracy reflects the wisdom of crowds. We find instead that it reflects the wisdom of a skilled minority.

2 Data

Prediction markets are organized around two key concepts: events and markets. An event groups one or more markets around a common question, with each market representing a binary Yes-or-No contract on a specific outcome. For example, the event “Fed decision in April (Apr 29)” contains multiple markets, one for each possible rate decision: a 50+ basis-point decrease, a 25 basis-point decrease, no change, or a 25+ basis-point increase. In contrast, the event “Will JPMorgan Chase (JPM) beat quarterly earnings on April 14, 2026?” has a single market with a Yes or No outcome.

For each market, participants can take a position on either side by buying Yes or No. Each contract pays \$1 if the corresponding outcome occurs and \$0 otherwise. If risk premia and time discounting are negligible, these prices approximate the physical probability of the event ([Wolfers and Zitzewitz, 2004, 2006](#); [Gomez Cram et al., 2026](#)).

Several platforms host prediction markets, providing the institutional setting in which these contracts are traded. Among them, Polymarket and Kalshi have emerged as the two largest by trading volume. We focus on Polymarket, the leading platform globally. Polymarket is built on Polygon, a public blockchain that records all transactions in a publicly verifiable and tamper-proof ledger. On this infrastructure, each Yes-or-No contract is repre-

sented as an outcome token, a unique digital asset tied to a specific market outcome. Trading occurs through a Central Limit Order Book (CLOB), the standard exchange mechanism in which participants post orders at specified prices and trades execute when buy and sell orders are matched.

Once an event concludes, its outcome is confirmed through the Universal Market Access (UMA) Protocol, a decentralized verification system in which an independent network of token holders votes on the result. If disputed, a formal challenge process is triggered before final settlement. There are no trading fees on Polymarket during our sample period.

Our data covers all Polymarket markets created after January 1, 2023, and resolved by December 31, 2025. Because the platform’s decentralized architecture stores market metadata, price histories, and transaction records in separate systems, we combine three data sources linked by a common token identifier. From the GAMMA API, we obtain market-level metadata, including event descriptions, categories, resolution dates, and token identifiers. From the CLOB API, we obtain historical price series for each outcome token. From the SubGraph, an indexing layer built on the blockchain, we obtain the complete record of individual transactions, including token identifier, trader address, trade size, price, direction, and timestamp.

Our final sample covers 98,906 events and 210,322 markets, with an average of 2.1 markets per event. The events span a wide range of topics, including politics, sports, cryptocurrency, finance, economics, culture, and technology, reflecting the broad scope of outcomes on which participants can trade. In total, the sample includes \$13.76 billion in trading volume across 1.72 million accounts. Table E.8 in the Appendix reports aggregate summary statistics, and Table E.9 provides a breakdown by topic.

Figure 1 documents the rapid growth of Polymarket and the distribution of trading activity across accounts. Panel A shows that monthly trading volume grew from \$3.3 million in December 2023 to \$1.98 billion in December 2025, a nearly 600-fold increase over two years. Panel B shows that the number of active accounts expanded in parallel, from approximately 1,600 to over 519,000 over the same period. Panel C reveals that trading activity is highly skewed across accounts: in December 2025, the median active account traded \$72, while the 99th percentile account traded \$74,000, a ratio exceeding 1,000 to one.

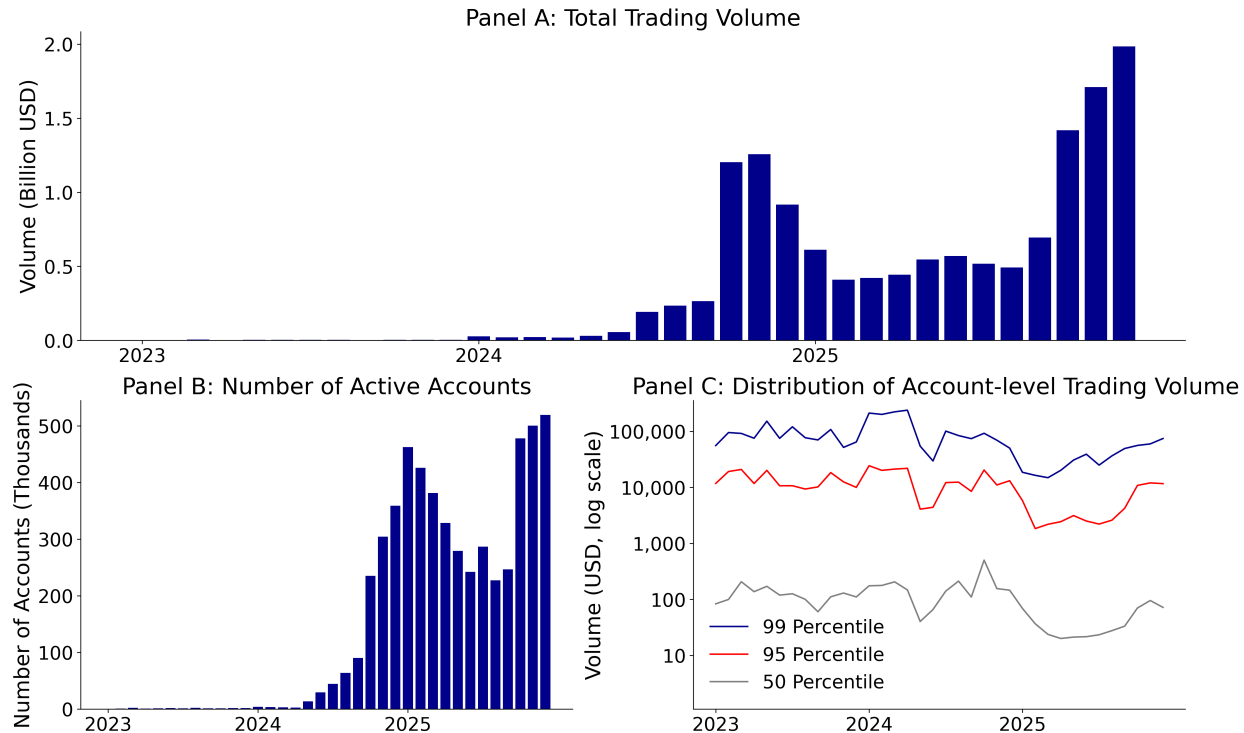


Figure 1: **Polymarket trading activity.** Panel A plots monthly trading volume in billions of U.S. dollars. Panel B plots the number of active accounts, where an account is active in a month if it executes at least one trade. Panel C plots the 50th, 95th, and 99th percentiles of monthly trading volume across active accounts, in U.S. dollars, on a log scale. The sample period is January 2023 to December 2025.

3 A Skilled Minority Drive Price Discovery

At the heart of our analysis is whether prediction markets are accurate because of the many or the few. Appendix A formalizes the two possibilities. Under the wisdom of crowds, every trader is independently and equally informed, and prices are accurate only because individual errors cancel out, so no single trader consistently outperforms. Under the alternative, only a small, informed minority drives accuracy. Using a new measure of skill that identifies this minority, we test the model’s three predictions that separate the few from the crowd: (i) skill persists out of sample; (ii) price discovery concentrates in the minority, not the crowd; and (iii) the minority’s order flow predicts outcomes even after controlling for prices.

3.1 Identifying the Skilled

We identify skilled traders with a novel measure that exploits the unique granularity of the data, where every trade can be linked to the account that placed it. The key challenge is that skill is not directly observable. Realized profit is the natural proxy, but it reflects luck as well as skill, since with hundreds of thousands of accounts many traders earn large profits by chance. To separate the two, we benchmark each trader’s profit against what they would have earned by trading at random. A trader who significantly exceeds this benchmark is skilled.

Consider a trader active in events $e = 1, \dots, E$, and let $\mathcal{K}_e$ be their trades in event e . Each trade k has size q_k , price p_k , outcome $Y_k \in \{0, 1\}$, and direction $d_k \in \{-1, 1\}$, where $d_k = 1$ denotes a buy and $d_k = -1$ denotes a sell. The trader’s realized profit and loss (PnL) is

$$\text{PnL} = \sum_{e=1}^E \sum_{k \in \mathcal{K}_e} q_k d_k (Y_k - p_k). \quad (1)$$

We generate the distribution of PnL under a null in which the trader’s direction in each event is random. We assign each event an independent sign $\varepsilon_e^{(b)} \in \{-1, 1\}$ with equal probability and apply it to all of that event’s trades, giving the simulated PnL

$$\text{PnL}^{(b)} = \sum_{e=1}^E \varepsilon_e^{(b)} \sum_{k \in \mathcal{K}_e} q_k d_k (Y_k - p_k), \quad (2)$$

with $b = 1, \dots, B$. We randomize at the event level, treating all of a trader’s trades in an event as a single decision. The level of randomization matters because it determines how many independent decisions the null attributes to a trader. We could instead randomize trade by trade, but this would overstate that number, splitting the many trades used to build a single position into separate bets. We could also randomize market by market, but this would overstate it too, splitting correlated positions in the same event into separate bets, such as “Trump wins” and “Harris loses” in 2024, which are the same bet since Harris loses exactly when Trump wins.

For each trader, we then compute the fraction of simulated PnLs that exceed their realized

PnL, plus half of those that equal it. Formally, the p -value is

$$p\text{-value} = \frac{\#\{\text{PnL}^{(b)} > \text{PnL}\} + \frac{1}{2}\#\{\text{PnL}^{(b)} = \text{PnL}\}}{B}, \quad (3)$$

where $\#\{\text{PnL}^{(b)} > \text{PnL}\}$ and $\#\{\text{PnL}^{(b)} = \text{PnL}\}$ count the simulated PnLs above and equal to the realized PnL, respectively. A trader active in E events has at most 2^E distinct simulated PnLs, one per sign pattern, and this discreteness has two implications. First, for small E ties with the realized PnL are common, which motivates the mid- p adjustment (Lancaster, 1961; Routledge, 1994); counting each tie in full would inflate the numerator and bias the p -value upward, against identifying the skilled. The half-count also makes the test symmetric, so a trader and the exact opposite who earns $-\text{PnL}$ receive complementary p -values, p and $1 - p$. Second, the same discreteness confines the p -value to $[2^{-(E+1)}, 1 - 2^{-(E+1)}]$; a realized value outside this range reflects only simulation noise, so we truncate at these trader-specific bounds.

We classify accounts into five groups. The p -value sorts the first four. An account is *skilled* if its p -value is below 0.05, so its PnL exceeds nearly all of the simulated outcomes. Between 0.05 and 0.95, an account performs no differently from random trading, so its gains or losses reflect luck. We label it *lucky* if its PnL is positive and *unlucky* if it is negative. Above 0.95, an account is *anti-skilled*, doing worse than random. The fifth group, *market makers*, we set apart by behavior rather than p -value, since their role is to provide liquidity, not to bet on outcomes. We classify an account as a market maker if (i) it trades in at least 100 markets, (ii) draws at least 70% of its volume from limit orders, and (iii) trades both sides of an outcome in at least 70% of the markets it trades. Because this rule relies only on behavior, we apply it first, removing market makers before classifying the remaining accounts by their p -value.

Figure 2 plots the distribution of p -values from the sign-randomization test across all non-market-maker accounts active in at least ten events. We use $B = 1,000$ simulations for each account. The horizontal dashed line marks the uniform density expected under the null of no skill, and the data depart from it sharply in both tails. On the left, a dark-blue spike collects the accounts we classify as skilled, those with a p -value below 0.05. On the

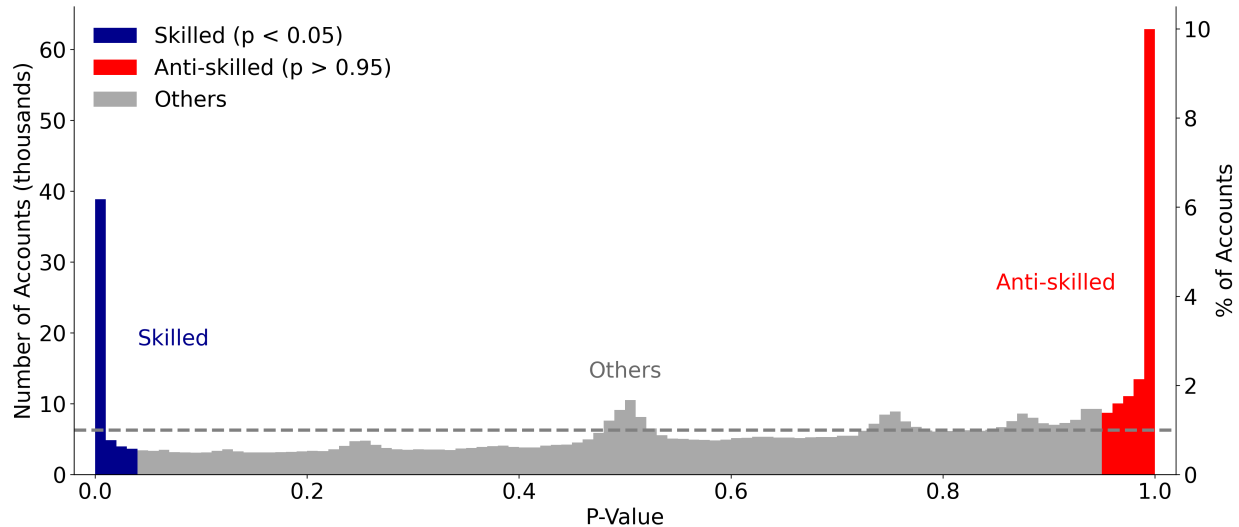


Figure 2: **Distribution of p -values from skill test.** p -values are obtained by comparing each account’s realized PnL to 1,000 simulated PnL outcomes under randomized trading directions at the event level. Dark blue bars indicate skilled accounts (p -value < 0.05); red bars indicate anti-skilled accounts (p -value > 0.95); gray bars indicate the remaining accounts. The horizontal dashed line marks the density corresponding to the uniform distribution expected under the null of no skill. The left axis reports accounts in thousands; the right axis reports the share of total accounts. Only accounts that trade in at least 10 events are included. The sample covers markets created after January 2023 and resolved by December 2025.

right, a red spike collects the anti-skilled, with a p -value above 0.95. Between them lies a broad, flat mass of accounts whose realized PnL cannot be told apart from random sign-flips, the luck-driven middle that makes up most of the cross-section. The first row of Table 1, Panel A, reports the counts. The skilled number 54,325 accounts, 8.63% of those active in at least ten events and 3.16% of all accounts. The anti-skilled number 105,951, or 16.84% and 6.15%, roughly twice as many as the skilled.

This classification of the skilled and the other groups is the one we use throughout the main text. As with any classification, it rests on assumptions, and we now show that they are not consequential.

First, we require at least ten events to classify an account as skilled or anti-skilled. Ten events give the test enough power to distinguish skill from luck and make the p -value approximately uniform under the null of no skill.⁴ A reasonable objection is that accounts

⁴With E events the randomization produces at most 2^E distinct simulated PnLs, so the test resolves a

Table 1: The Share of Skilled and Anti-Skilled Across Specifications

Event filter	Skilled			Anti-skilled		
	Number	As percentage of		Number	As percentage of	
		Filtered	All		Filtered	All
Panel A: No Multiple-testing Adjustment						
≥ 10 events (benchmark)	54,325	8.63	3.16	105,951	16.84	6.15
≥ 4 events	82,186	7.70	4.77	212,726	19.94	12.36
None	82,186	4.77	4.77	212,726	12.36	12.36
Panel B: Benjamini–Hochberg Adjustment						
≥ 10 events	33,522	5.33	1.95	49,590	7.88	2.88
≥ 4 events	33,442	3.13	1.94	53,122	4.98	3.09
None	27,422	1.59	1.59	38,793	2.25	2.25

Notes: The table reports the number of accounts classified as skilled ($p < 0.05$) and anti-skilled ($p > 0.95$) under the sign-randomization test, with each count as a share of two denominators across event-count filters. *Filtered* is the set of accounts meeting the event filter in that row; *All* is the full set of accounts. Panel A applies no multiple-testing adjustment; Panel B controls the false discovery rate at the 0.05 level using the Benjamini–Hochberg procedure (Benjamini and Hochberg, 1995), applied separately to each tail. The benchmark specification used throughout the paper is ≥ 10 events with no adjustment (first row of Panel A). Because each p -value is bounded in $[2^{-(E+1)}, 1 - 2^{-(E+1)}]$, accounts active in fewer than four events can never reach $p < 0.05$ or $p > 0.95$; removing the filter (“None”) therefore leaves the classified counts identical to the ≥ 4 filter and changes only the denominator.

trading in fewer than ten events are never given the chance to be significant, since the filter excludes them from the test universe. We therefore report the skilled share under other event filters in Table 1, Panel A. Lowering the threshold to four events, or removing it altogether, leaves the skilled at 4.77% of all accounts and the anti-skilled at 12.36%.⁵

A second concern is that we apply the test separately to many accounts, which raises multiple-testing issues (Harvey, Liu, and Zhu, 2016; Harvey, 2017). Under the null that no account has skill, about 5% would be flagged as skilled and 5% as anti-skilled by chance alone. We therefore also report shares that control the false discovery rate by the Benjamini–

p -value only as finely as the smaller of 2^E and B ; ten events lift 2^E to $2^{10} = 1,024$, just past our 1,000 draws, so the event count no longer limits resolution and the null is approximately uniform.

⁵Because the p -value is bounded below by $2^{-(E+1)}$, four events is the fewest at which a value below 0.05 is reachable, so accounts with fewer can never be skilled, and dropping the filter changes only the denominator, not the set of skilled accounts.

Hochberg procedure (Benjamini and Hochberg, 1995), in Table 1, Panel B. The correction lowers these shares but does not eliminate them. For example, for accounts trading ten events or more, the fraction of skilled traders within the filtered universe drops to 5.33%, which amounts to 1.95% of the full universe. Lowering the threshold to 4 events or more, the comparable shares are 3.13% and 1.94%, respectively.

In Appendix B we show that none of these assumptions, nor the other choices behind the classification, drives our main results. Specifically, we vary the event-count filter and the multiple-testing adjustment, replace the sign-randomization benchmark with one that also randomizes trade sizes and with one that resamples market outcomes from their prices, and vary the criteria that identify market makers. Under each of these classifications, the results are the same as those we establish for our main specification in the next sections, namely that the skilled group’s order flow predicts future prices and final outcomes and reduces pricing errors, while the other groups’ does not.

3.2 Trading Activity and Performance

Having classified accounts into five groups, we now examine how they trade and perform. Table 2, Panel A reports each group’s accounts, PnL, and volume. The most striking pattern is the concentration of gains. Market makers and the skilled together represent fewer than 3.5% of all accounts (Column 2), yet capture over 30% of total gains (Column 4). The lucky, a larger group at 29% of accounts, capture the remaining 70%. On the losing side, the unlucky and the anti-skilled make up 68% of all accounts and absorb the entirety of aggregate losses. A minority of participants thus earn all of the gains, while the majority lose. The gains of market makers, the skilled, and the lucky are fully offset by the losses of the unlucky and the anti-skilled, consistent with the market’s zero-sum arithmetic.

Panel B reports account-level averages within each group. Market makers operate at a distinct scale, trading across 942 markets on average (Column 2) and earning \$11,832 per account (Column 3). Among the remaining groups, skilled traders are notably more active than the lucky. They trade in 78 markets on average, more than three times as many as the lucky (23 markets), suggesting their ability is not limited to a small number of markets. They also earn \$3,243 per account compared to \$914 for the lucky. The unlucky and the anti-

Table 2: Trading Activity and Performance by Account Group

Panel A: Group-level Totals						
	Account		PnL		Volume	
	Number	Share	Amount	Share	Amount	Share
	(Thousand)	(%)	(Million \$)	(%)	(Billion \$)	(%)
	(1)	(2)	(3)	(4)	(5)	(6)
Market Maker	1.66	0.10	19.58	3.01	2.18	7.93
Skilled	54.33	3.16	176.20	27.07	6.27	22.75
Lucky	498.13	28.93	455.10	69.92	8.34	30.27
Unlucky	1,061.58	61.66	−531.82	−81.70	9.17	33.30
Anti-skilled	105.95	6.15	−119.09	−18.30	1.56	5.66
Panel B: Account-level Averages Within Group						
	Number of	Number of	PnL	Volume	Top 1 Event	Profit
	Events	Markets	(\$)	(\$)	Profit Share (%)	HHI
	(1)	(2)	(3)	(4)	(5)	(6)
Market Maker	662.69	942.37	11,831.93	1,320,107.38	13.62	0.06
Skilled	60.22	78.35	3,243.42	115,391.76	34.58	0.24
Lucky	18.88	22.75	913.61	16,741.48	71.00	0.64
Unlucky	14.29	17.70	−500.97	8,641.80	62.49	0.55
Anti-skilled	40.21	52.20	−1,123.97	14,719.42	55.65	0.47

Notes: This table reports trading activity and performance by trader group. Panel A reports group-level totals. **Account Number (Share)** is the number (share) of unique accounts in each group, in thousands (percentage). **PnL Amount** is aggregate profit and loss in millions of USD. **PnL Share** is the group’s aggregate gains (losses) as a share of total gains (losses) across all groups. **Volume Amount (Share)** is the amount (share) of aggregate trading volume in billions of USD (percentage). Panel B reports account-level averages within each group. **Number of Events (Markets)** is the average number of events (markets) traded per account. **PnL** is average profit and loss per account in USD. **Volume** is the average trading volume per account in USD. **Top 1 Event Profit Share** is the share of an account’s total profits attributable to its single most profitable event. **Profit HHI** is the Herfindahl–Hirschman Index computed over the shares of each account’s profit-generating events.

skilled lose \$501 and \$1,124 per account, respectively. Table E.10 in the appendix provides the full distribution of PnL within each group.

Columns 5 and 6 of Panel B measure the concentration of each group’s profits across events. Column 5 reports the share of total profits attributable to an account’s single most profitable event. For market makers, this share averages 14%; for skilled traders, 35%. The lucky are far more concentrated at 71%, and the unlucky and the anti-skilled concentrate 62% and 56% of their profits in a single event. The Profit Herfindahl–Hirschman Index

(HHI) in Column 6 confirms this pattern. Most participants’ gains hinge on a small number of outcomes; skilled traders and market makers, by contrast, earn profits consistently across events. Table E.11 in the appendix reports additional concentration measures, including the effective number of profit-generating trades and the rank-weighted profit index.

We refer to the lucky, the unlucky, and the anti-skilled collectively as the crowd. Together they make up roughly 97% of accounts and supply about 70% of trading volume.

3.3 Skill Is Persistent, Luck Is Not

The fact that skilled traders account for only 3% of all accounts is our main evidence against the wisdom-of-crowds explanation for prediction market accuracy. But the evidence rests on the appropriateness of our statistical measure of skill. We therefore validate the measure empirically via an out-of-sample test. If skill is genuine and well-measured, it should persist out-of-sample.

We randomly split events into training and test sets, classify accounts in the training set, and evaluate whether the classification holds in the test set. We split events randomly rather than by time because market conditions, including the user base, event types, and liquidity, vary substantially over the sample period, particularly in late 2025 with a wave of new users and markets (as shown in Figure 1). A random split ensures that both sets contain events throughout the full sample period and share a comparable user base.

Table 3, Panel A reports the correlation of account-level PnL ranks and p -values between the training and test sets. We first examine PnL. The rank correlation is 0.11 for the full sample (Column 1), but this reflects the persistence of losses (0.12, Column 3), not gains. In fact, training-set winners tend to reverse, with a correlation of -0.12 (Column 2). High realized profits are, therefore, a poor predictor of future performance.

Our statistical measure of skill is much more persistent. The correlation of sign-randomized p -values between the training and test sets is 0.35 (Column 4), over three times the PnL rank correlation. This holds for both the skilled subsample (0.27, Column 5) and the non-skilled subsample (0.25, Column 6), meaning that skill persists, as does the absence of skill. Traders identified as skilled in the training set remain skilled out of sample, validating the sign-randomization-based classification as a measure of persistent ability rather than luck.

Panel B reinforces this finding with a transition matrix of group classifications between the training and test sets. Market makers are the most persistent: 84% retain their classification. Among skilled traders, 46% remain skilled, and 22% become lucky, so nearly 70% remain profitable. The boundary between skilled and lucky is inherently noisy, as some skilled traders may lack sufficient data in the test set to be identified, and some of the lucky may possess genuine skill that our test does not detect. The persistence of “anti-skill” is similarly high, with 48% of the anti-skilled from the training set retaining their classification in the test set.

The persistence of skill and anti-skill in prediction markets appears unusually high compared with that documented in other financial markets. To make this comparison explicit, we replicate our test with active mutual fund managers in Appendix D. Specifically, we randomly split their return history into training and test sets and classify funds into four groups based on the sign and significance of their alpha from a regression of fund gross returns on the four factors in the model of [Carhart \(1997\)](#). Only 10% of skilled mutual funds in the training set retain their classification in the test set, whereas the corresponding number for anti-skilled mutual funds is 6%. These numbers are much lower than in prediction markets. Thus, relative to mutual funds, prediction markets provide stronger evidence that some traders can consistently beat the market.

Luck is not persistent. For example, the table shows that 64% of the lucky in the training set become losers in the test set. An illustrative case is the account @majorexploder, classified as lucky after betting \$4.5 million on three sports matches over one weekend in early 2026 and winning \$3.6 million. Such concentrated gains are unlikely to persist, and most of the lucky end up on the losing side in the subsequent sample. Overall, the classification based on skill is stable out of sample, whereas the classifications based on realized profits are not.

Just as the skilled persistently win, the anti-skilled persistently lose. This raises the question: why do these accounts lose systematically rather than simply break even? One explanation is a predictable behavioral mistake: in Section 4.3.3 we show that they overpay for very unlikely outcomes, losing steadily where random trading would break even. Harder to explain is why they keep trading at such a loss; one possibility is that they derive utility

Table 3: Persistence of Skill and Profits/Losses

Panel A: Correlation of Profit and Skill Measures between the Training Set and the Test Set						
	Variable: PnL Rank			Variable: p -value		
	Full	Restricted Sample		Full	Restricted Sample	
	Sample	Profit	Loss	Sample	Skill	Non-skill
	(1)	$\text{PnL}^{\text{train}} > 0$	$\text{PnL}^{\text{train}} < 0$	(4)	$p^{\text{train}} < 0.5$	$p^{\text{train}} > 0.5$
		(2)	(3)		(5)	(6)
Correlation	0.11	-0.12	0.12	0.35	0.27	0.25
Observations	1,290,985	439,047	834,927	1,290,985	391,567	895,792

Panel B: Transition Matrix from the Training Set to the Test Set					
Training Set	Test Set				
	Market Maker	Skilled	Lucky	Unlucky	Anti-skilled
Market Maker	0.84	0.03	0.05	0.06	0.01
Skilled	0.00	0.46	0.22	0.30	0.01
Lucky	0.00	0.03	0.32	0.61	0.03
Unlucky	0.00	0.02	0.23	0.73	0.03
Anti-skilled	0.00	0.01	0.18	0.33	0.48

Notes: The table shows the persistence of skill and profits and losses (PnL). Panel A tests whether in-sample PnL rank and skill p -value persist out of sample. Columns 1–3 report the correlation between accounts’ normalized PnL rank in the training set and in the test set. Column 1 uses the full sample; column 2 restricts the sample to accounts that make a profit in the training sample ($\text{PnL}^{\text{train}} > 0$) and column 3 to accounts that lose in the training sample ($\text{PnL}^{\text{train}} < 0$). Columns 4–6 report the correlation between accounts’ p -value, constructed from our sign-randomization test, in the training set and in the test set. Column 4 uses the full sample; column 5 restricts to accounts with a p -value below 0.5 in the training sample and column 6 restricts to accounts with a p -value above 0.5 in the training sample. Panel B reports the transition matrix of account group classifications between training and test sets. Each row gives the fraction of training-set group members assigned to each test-set group, so each row sums to one.

from betting itself.

3.4 Skilled Trades Predict Outcomes

Having identified a small group of persistently skilled traders, we now ask whether these traders are the main reason why prediction markets are accurate, or whether the accuracy comes from broad participation across multiple groups. We first show that skilled traders’ order flow predicts both future price movements and final outcomes, whereas the crowd’s

does not.

To establish this, we construct a group-level measure of order imbalance. For each market, we divide its duration into 50 equal-length time bins, so that markets with different lifespans are comparable. The median market in our sample lasts 2.36 days, so for such a market, each bin spans approximately 1.13 hours. Within each bin, we compute the total share volume of buys and sells by each group and use these to construct a measure of order imbalance. For group g , market m , and time bin t , we define

$$\text{OIB}_{m,t}^g = \frac{B_{m,t}^g - S_{m,t}^g}{B_{m,t}^g + S_{m,t}^g}, \quad (4)$$

where $B_{m,t}^g$ and $S_{m,t}^g$ are the total share buy and sell volume by group g , and $B_{m,t} + S_{m,t}$ is total volume across all groups. The measure ranges from -0.5 to 0.5 : it equals 0.5 when group g is the only buyer and -0.5 when it is the only seller.

We run two regressions to test whether a group’s trading activity predicts prices. In both, we regress an outcome on current order imbalance for each group g ,

$$y_{m,t} = \alpha^g + \beta^g \cdot \text{OIB}_{m,t}^g + \text{Controls}_{m,t-1} + \varepsilon_{m,t}, \quad (5)$$

where the outcome $y_{m,t}$ is either the next-period price change $\Delta p_{m,t+1}$ or the final market outcome p_m^{outcome} . A positive β^g indicates that group g ’s net buying predicts higher future prices or a higher probability of the event occurring. The controls are lagged price, lagged price change, lagged log of trading volume, and market duration. Accounts are classified on the training set; all results are reported on the test set.

Table 4 reports the results. Panel A uses the next-period price change as the dependent variable; Panel B uses the final market outcome. Skilled traders’ order imbalance is a strong predictor of both: the coefficient is 2.05 in Panel A ($t = 12.81$) and 7.77 in Panel B ($t = 9.57$). A one percentage point increase in skilled OIB is associated with a 2 basis point increase in the next-period price and an 8 basis point increase in the probability of the final outcome. The lucky, despite earning positive profits, show no significant predictive power in either specification ($t = 1.45$ in Panel A and $t = 1.45$ in Panel B). Only skilled traders’ order imbalance predicts future prices; positive profits alone do not imply informational content.

Table 4: Regressions of Price Changes and Outcomes on Order Imbalance

Panel A: Dependent Variable $\Delta p_{m,t+1}$					
	Skilled (1)	Lucky (2)	Unlucky (3)	Anti-skilled (4)	Market Maker (5)
OIB	2.05	0.18	-1.78	-2.52	1.26
<i>t-statistic</i>	[12.81]	[1.45]	[-24.69]	[-11.12]	[12.49]
Observations	1,267,183	1,267,183	1,267,183	1,267,183	1,267,183
R^2	0.001	0.000	0.001	0.001	0.001
Panel B: Dependent Variable p_m^{outcome}					
	Skilled (1)	Lucky (2)	Unlucky (3)	Anti-skilled (4)	Market Maker (5)
OIB	7.77	0.74	-0.37	-9.74	-5.69
<i>t-statistic</i>	[9.57]	[1.45]	[-0.66]	[-6.77]	[-10.31]
Observations	1,362,338	1,362,338	1,362,338	1,362,338	1,362,338
R^2	0.327	0.325	0.325	0.326	0.326

Notes: This table reports regressions of subsequent price changes and final market outcomes on each account group’s order imbalance (OIB). OIB is defined as the net buy volume of the group scaled by total trading volume within market-normalized time bins. Panel A uses the next-period price change as the dependent variable; Panel B uses the final market outcome. Accounts are classified based on the training set; all results are reported on the test set. The controls are lagged price, lagged price change, lagged log trading volume, and market duration. *t*-statistics are shown in brackets, with standard errors clustered at the event and month levels.

The remaining groups display patterns consistent with their economic roles. Market makers’ OIB positively predicts next-period price changes (1.26, $t = 12.49$) but negatively predicts the final outcome (-5.69 , $t = -10.31$), consistent with their accommodating informed order flow in the short run and ending up on the wrong side at resolution. The unlucky negatively predict next-period prices (-1.78 , $t = -24.69$) but have no significant relationship with final outcomes. The anti-skilled negatively predict both next-period prices (-2.52 , $t = -11.12$) and final outcomes (-9.74 , $t = -6.77$), trading consistently in the wrong direction. Across all groups, only the skilled predict both future prices and final outcomes, while the rest, the crowd that supplies most of the volume, predict nothing or trade against the eventual outcome. Information comes from the skilled few, not the many.

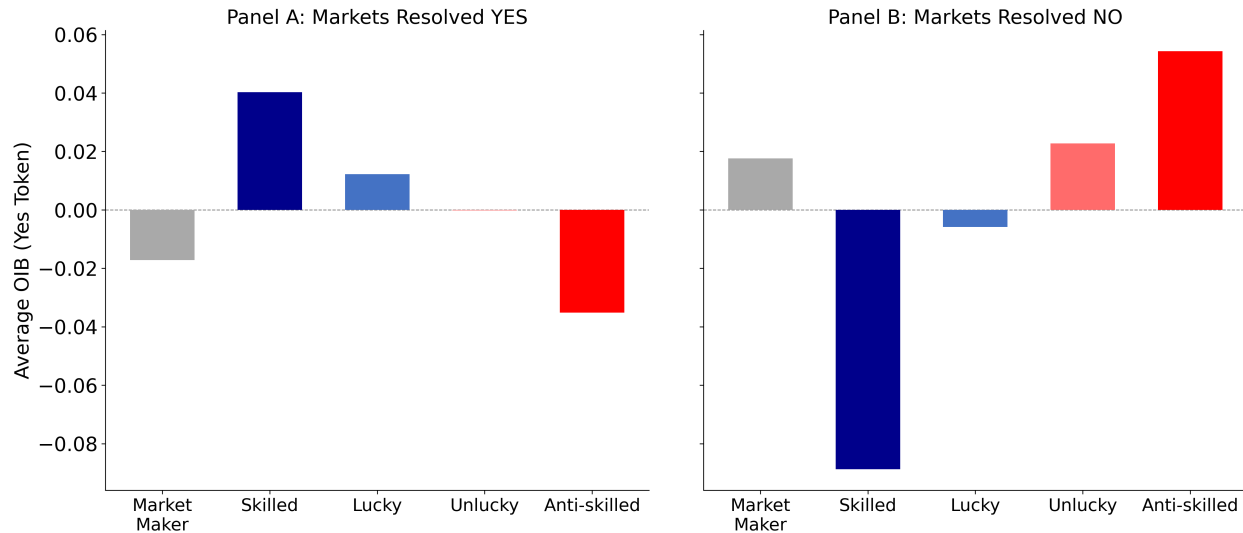


Figure 3: **Order imbalance before resolution by group.** Each bar plots the average order imbalance (OIB) in the last 5% of each market's lifespan by account group. Panel A shows markets that resolved Yes; Panel B shows markets that resolved No. OIB is the net buy volume for the Yes outcome divided by total volume within the period, so that positive (negative) values indicate net buying (selling) of the Yes outcome. Accounts are classified based on the training set; all results are reported on the test set.

The regression analysis pools trading activity across all time bins within a market's life. However, approximately 43% of total trading volume is concentrated in the last 5% of each market's lifespan. We therefore focus on this period, when prices should most closely reflect the final outcome. Figure 3 plots the average OIB by group in bar chart form, where Panel A shows markets that resolved Yes and Panel B shows markets that resolved No.

The pattern is striking. In markets that resolved Yes, skilled traders are net buyers; in markets that resolved No, they are net sellers. Skilled traders consistently position in the direction of the final outcome. The lucky follow the same direction but with substantially lower magnitude, consistent with their weaker informational content. On the opposite side, the unlucky and the anti-skilled consistently position against the final outcome, with the anti-skilled showing the larger magnitude of the two. Market makers are slightly net sellers in markets that resolved Yes and net buyers in markets that resolved No, consistent with accommodating directional order flow and earning the bid-ask spread rather than taking informed positions.

3.5 Skilled Traders Improve Accuracy

Prediction markets are remarkably accurate. But accuracy is an outcome: someone’s trades must actually move prices toward the final outcome. The previous section showed that skilled traders’ trades predict future prices and final outcomes, while other groups’ do not. This section asks whether those predictive trades are also the ones that make prices more accurate throughout the lifespan of the event. For each time window, we measure how much the price moves toward or away from the final outcome, and test which groups’ contemporaneous trades drive this movement.

We define the price discovery contribution over a narrow time window between $t - 1$ and t as the change in absolute pricing error:

$$PD_{m,t-1,t} = |p_{m,t} - p_m^{\text{outcome}}| - |p_{m,t-1} - p_m^{\text{outcome}}|, \quad (6)$$

where $p_m^{\text{outcome}} \in \{0, 1\}$ is the final outcome of market m . A negative value indicates that the price moved closer to the final outcome and became more accurate; a positive value indicates that the price moved away and became less accurate. To make markets with different durations comparable, we standardize each market’s lifespan to the unit interval and divide it into 50 equal-length bins, as in the previous section. The index t denotes the end of a bin and $t - 1$ its start, so the window from $t - 1$ to t corresponds to one bin. This gives us one observation of price accuracy change per market-bin.

To test which groups drive these changes, we measure each group’s share of trading activity in the same window. For group g , market m , and time bin t , we define the group-level volume share as

$$\text{VOL}_{m,t}^g = \frac{B_{m,t}^g + S_{m,t}^g}{B_{m,t} + S_{m,t}}, \quad (7)$$

where $B_{m,t}^g$ and $S_{m,t}^g$ are the total share buy and sell volume by group g , and $B_{m,t} + S_{m,t}$ is total volume across all groups. The measure ranges from 0 to 1: it equals 1 when group g is the only trader in the window and 0 when the group does not trade. The measure is unsigned because $PD_{m,t-1,t}$ already reflects whether the price moved toward or away from

the outcome. A negative coefficient on $\text{VOL}_{m,t}^g$ therefore means a group’s trading improves accuracy on average.

We estimate the panel regression for each group g :

$$PD_{m,t-1,t} = \alpha^g + \beta^g \cdot \text{VOL}_{m,t}^g + \text{Controls}_{m,t-1} + \varepsilon_{m,t}. \quad (8)$$

The coefficient of interest, β^g , measures how a group’s trading activity in a window translates into price accuracy: a one percentage point increase in group g ’s volume share is associated with a β^g change in the bin’s price error. Controls are the same as in the previous section (lagged price, lagged price change, lagged log trading volume, and market duration). We cluster standard errors at the event level and at the calendar month containing the midpoint of the bin. Event-level clustering accounts for correlation across markets and time bins within the same event, while month-level clustering accounts for correlation among contemporaneous markets from platform-wide shocks.

Table 5 reports the results. Panel A pools all time bins across the lifespan of each market. Only skilled traders’ volume share carries a negative and statistically significant coefficient: a one percentage point increase in skilled VOL is associated with a 5 basis point reduction in the bin’s price error ($t = -5.54$). The coefficients for the crowd (the lucky, the unlucky, and the anti-skilled) are all positive, and significant for the lucky and the unlucky, meaning their trading, on average, moves prices away from the outcome and makes them less accurate. Market makers’ coefficient is negative but not significant, consistent with their role as liquidity providers rather than information producers.

Panel B splits the sample by the bin’s position in the market’s lifespan, from the first 20% through the final 20%. The negative and significant coefficient on skilled VOL holds in every window and grows substantially larger toward resolution, from -0.59 in the first 20% to -9.53 in the final 20%. This pattern is consistent with Figure 3: skilled traders contribute to price accuracy throughout the lifespan of a market, but most of the accuracy improvement is concentrated in the final bins as the outcome nears resolution. Skilled traders are the ones who make prices more accurate; the crowd, despite supplying most of the volume, moves them away from the outcome.

Table 5: Regressions of Absolute Price Error on Trading Activity

Panel A: Full Sample					
	Skilled (1)	Lucky (2)	Unlucky (3)	Anti-skilled (4)	Market Maker (5)
VOL	−4.95	1.64	2.52	0.70	−0.67
<i>t</i> -statistic	[−5.54]	[2.95]	[6.48]	[1.55]	[−1.37]
Observations	1,265,693	1,265,693	1,265,693	1,265,693	1,265,693
R^2	0.031	0.024	0.025	0.022	0.022
Panel B: Decomposition by Market Lifespan					
	Skilled (1)	Lucky (2)	Unlucky (3)	Anti-skilled (4)	Market Maker (5)
<i>First 20% of Market Lifespan</i>					
VOL	−0.59	0.25	0.39	0.58	−0.41
<i>t</i> -statistic	[−5.06]	[2.06]	[3.78]	[3.06]	[−4.17]
Observations	95,786	95,786	95,786	95,786	95,786
R^2	0.006	0.006	0.006	0.006	0.006
<i>20–40% of Market Lifespan</i>					
VOL	−0.72	−0.05	0.40	0.69	0.05
<i>t</i> -statistic	[−7.35]	[−0.96]	[5.39]	[4.19]	[0.75]
Observations	175,379	175,379	175,379	175,379	175,379
R^2	0.004	0.004	0.004	0.004	0.004
<i>40–60% of Market Lifespan</i>					
VOL	−1.17	−0.01	0.56	0.76	0.21
<i>t</i> -statistic	[−5.34]	[−0.05]	[7.27]	[6.07]	[1.40]
Observations	230,522	230,522	230,522	230,522	230,522
R^2	0.003	0.002	0.002	0.002	0.002
<i>60–80% of Market Lifespan</i>					
VOL	−1.91	0.13	0.88	0.94	0.29
<i>t</i> -statistic	[−6.16]	[1.15]	[8.25]	[3.94]	[1.40]
Observations	302,303	302,303	302,303	302,303	302,303
R^2	0.005	0.003	0.003	0.003	0.003
<i>Final 20% of Market Lifespan</i>					
VOL	−9.53	3.36	5.28	2.29	−2.58
<i>t</i> -statistic	[−7.91]	[3.60]	[8.61]	[2.14]	[−1.88]
Observations	470,327	470,327	470,327	470,327	470,327
R^2	0.037	0.023	0.027	0.021	0.021

Notes: This table reports regressions of the change in absolute pricing error ($PD_{m,t-1,t}$) on each account group's scaled trading volume (VOL). VOL is defined as the trading volume of the group scaled by total trading volume within market-normalized time bins. Panel A pools all observations across the full market lifespan. Panel B reports the same regression separately for five equal windows of the market lifespan. Accounts are classified based on the training set; all results are reported on the test set. The controls are lagged price, lagged price change, lagged log trading volume, and lagged log market duration. *t*-statistics are shown in brackets, with standard errors clustered at the event and month levels.

4 The Skilled Minority Are Not Entirely Insiders

Section 3 showed that a small minority drives prediction market accuracy, which rules out the wisdom of crowds. But that same minority could be insiders trading on private information rather than skilled traders processing public information.⁶ We now show that it is the latter. No single test can rule out insider trading entirely, so we build a cumulative case from three pieces of evidence. First, skilled traders drive price discovery even in markets that are not prone to insider trading. Second, direct proxies identify only a few insiders, concentrated in a small number of markets. Third, skilled traders trade on public information. Taken together, these results lead us to conclude that insider trading is not a major contributor to prediction market accuracy.

4.1 Price Discovery in Non-Insider-Prone Markets

Identifying insiders is inherently difficult, because insiders will do their best to avoid being caught, for example, by spreading their trades over multiple accounts. Identifying markets where insider trading is possible, however, is considerably easier.

For every market in our sample, we ask a Large Language Model (LLM) whether it is prone to insider trading, requesting a response of “yes,” “no,” or “unclear.” We use Claude Haiku and give it strict instructions in the form of a rubric, which states that a contract is insider-prone if its outcome can, in principle, be (1) controlled or (2) privately known by some individual at the time bets are placed. The prompt also includes 100 hand-labeled markets and worked examples explaining each classification. The full approach is described in Appendix C.2.

Certain categories, like Culture and Politics, are highly prone to insider trading, with more than 70% of markets classified by the LLM as “yes.” Markets in these categories tend to have outcomes that can be controlled or known by specific individuals, such as “Will Trump launch a coin before the election?,” which makes them susceptible to insider trading. The largest Politics markets, like the 2024 U.S. presidential election, are typically classified

⁶Insider trading is documented in other derivatives markets: [Acharya and Johnson \(2007\)](#) find that the credit default swap market reveals private information held by informed banks ahead of adverse credit events.

Table 6: Order Flow and Price Dynamics of the Skilled, by Market Type

	Markets Not Prone to Insiders			Markets Prone to Insiders		
	$\Delta p_{m,t+1}$	p_m^{outcome}	$PD_{m,t-1,t}$	$\Delta p_{m,t+1}$	p_m^{outcome}	$PD_{m,t-1,t}$
	(1)	(2)	(3)	(4)	(5)	(6)
OIB	1.90	7.49		2.73	8.49	
<i>t-statistic</i>	[12.72]	[7.62]		[8.17]	[13.39]	
VOL			-5.44			-2.42
<i>t-statistic</i>			[-5.22]			[-15.85]
Observations	940,389	1,023,960	938,852	326,859	338,377	326,841
R^2	0.001	0.291	0.037	0.003	0.441	0.008

Notes: This table reports regressions of outcomes on the trading activity of the skilled, estimated on markets not prone to insider information (columns 1–3) and markets prone to insider information (columns 4–6). The dependent variables are the next-period price change $\Delta p_{m,t+1}$, the realized outcome p_m^{outcome} , and the change in absolute pricing error $PD_{m,t-1,t}$. OIB is the standardized order imbalance of the skilled (net buy volume scaled by total volume within market-normalized time bins), used for $\Delta p_{m,t+1}$ and p_m^{outcome} ; VOL is their standardized trading volume, used for $PD_{m,t-1,t}$. Accounts are classified on the training set; all results are reported on the test set. All specifications use lagged price, lagged price change, lagged log volume, and lagged log duration as controls. *t*-statistics are in brackets, with standard errors clustered at the event and month levels.

as not prone to insider trading (“no”), since the outcome is decided by a diffuse electorate that no individual controls or knows in advance. The vast majority of Sports and Crypto markets are also classified as not prone, with “no” classifications for 98.6% and 98.7% of markets, respectively. Because Sports and Crypto account for a large fraction of the sample, the majority (89%) of markets are classified as not prone to insider trading.

To test whether the skilled are insiders, we re-estimate the price discovery regressions of Sections 3.4 and 3.5 separately in markets the LLM classifies as prone to insider trading (“yes” or “unclear”) and in those it does not (“no”). Table 6 shows that skilled traders drive price discovery in both the markets that are prone to insider trading and, importantly, in those that are not. If skilled traders relied solely on inside information, they would only drive price discovery in insider-prone markets. But since they also drive price discovery in markets where inside information is unavailable, it cannot be the full explanation. Furthermore, since markets not prone to insider trading account for roughly 90% of our sample, it can also not be the main explanation.

4.2 Insiders Are Few and Localized

To gauge how widespread insider trading is, we flag suspicious accounts using three criteria: an implausibly strong track record, timing, and conviction. Before defining them precisely, we illustrate why they are informative with a concrete case.

On January 3, 2026, the U.S. military carried out a secret operation to capture Venezuelan President Nicolás Maduro. In the days and hours before the raid, three Polymarket accounts took sizable positions in the contract on his removal and collectively realized profits of over \$620,000. On April 23, 2026, the Commodity Futures Trading Commission filed a complaint alleging that Van Dyke, an active-duty U.S. Army service member, had engaged in insider trading through one of these accounts. The trades show why our criteria are informative. All three accounts were created between December 27 and January 3, just before the raid, and placed orders totaling hundreds of thousands of shares in a contract the market priced near \$0.10. Two traded nothing but Maduro and fell silent once it resolved; the third placed one more bet before going quiet. Appendix C.1 describes the case in further detail.

Our first and preferred measure is based on a trader’s *record*. It captures four account-level characteristics that most insiders satisfy but that are very hard to satisfy on public information alone. We classify a trader as an insider if they (i) have a per-share win rate of at least 90%, (ii) earn at least 0.1 in profit per share, (iii) make at least \$1,000 in total profits, and (iv) earn at least 90% of those profits in insider-prone markets.⁷ This measure flags all three suspected Maduro insiders, as well as other prominent cases, such as the Google software engineer who traded contracts on the most-searched person on Google in 2025.

Our second measure is based on *timing*. It targets a trader who opens an account just before an insider-prone event, trades it profitably, and stops trading after. It is narrower than the record measure but arguably more precise. We classify a trader as an insider if they (i) open the account no more than seven days before the event, (ii) make no trades after it resolves, (iii) trade only that one insider-prone event, (iv) trade at least \$1,000, and (v) earn

⁷The combination of win rate and profit per share is important to rule out skilled non-insiders. A non-trivial fraction of skilled traders have a win rate above 90%, which they achieve by trading against the favorite-longshot bias, buying contracts priced near one and selling those priced near zero. This ensures a high win rate, but their profit per share is far below the 0.1 we require of insiders.

at least \$1,000 in profits. This measure flags two of the three suspected Maduro insiders.

Our third measure is based on *conviction*. It targets traders who trade insider-prone events as if they are insensitive to the price at which they trade. We call a trade price-insensitive if it moves the price by 0.2 or more. We classify a trader as an insider if they (i) make a price-insensitive trade in an insider-prone market that they go on to win, (ii) earn more than half of their profits in markets where they trade this way, and (iii) make at least \$1,000 in profits.

Table 7, Panel A, reports summary statistics for the accounts each proxy flags as insiders. The record measure flags 757 accounts, timing 297, and conviction 124; because the measures overlap, the number flagged by at least one of them is 967. These accounts are a small part of the sample: they trade in 6.0% of markets, earn 1.3% of profits, account for 0.2% of volume, and are active in just 7.5% of time bins. Insiders are thus localized and sporadic, not a pervasive feature of most markets. Moreover, only 11 of the 967 flagged accounts (1.1%) are classified as skilled, so the skilled minority and the insiders are almost entirely distinct groups.

Table 7, Panel B, repeats the price discovery tests from Sections 3.4 and 3.5, now with insiders as a separate group. Columns 1–3 report the insider estimates; columns 4–6 repeat the tests for skilled traders, with insiders removed. Where insiders trade, their effect is large. A one percentage point increase in their order imbalance moves the next-period price by 1.91 basis points ($t = 2.12$) and the final-outcome probability by 22.54 basis points ($t = 6.48$), while a one percentage point increase in their volume cuts pricing error by 8.79 basis points ($t = -6.13$). Each effect is larger than the skilled traders' own on the same markets (2.95, $t = 8.91$; 9.82, $t = 7.51$; and -2.99 , $t = -8.41$).

Insiders, then, have large effects where they trade: they move prices, predict outcomes, and improve accuracy more than skilled traders do. But these effects are local, confined to the few markets and brief windows in which insiders are active. Removing insiders barely changes the skilled estimates in columns 4–6, because insiders are under 1% of skilled traders. Prediction market accuracy is the work of skill, not of insiders, who are decisive in the few markets they touch but far too rare to move prices market-wide.

Table 7: Insider Accounts: Summary Statistics and Price Impact

Panel A: Summary Statistics by Insider Type						
	Number of Accounts (1)	Share of Markets (%) (2)	Share of Total Profits (%) (3)	Share of Total Volume (%) (4)	Share Classified as Skilled (%) (5)	Share of Active Time Bins (%) (6)
Record	757	0.5	1.1	0.1	1.2	0.7
Timing	297	0.1	0.4	0.0	0.0	0.2
Conviction	124	5.7	0.1	0.1	1.6	6.8
All Insiders	967	6.0	1.3	0.2	1.1	7.5
Panel B: Order Flow and Price Dynamics (Restricted to Insiders-traded Markets)						
	Insiders			Skilled excluding Insiders		
	$\Delta p_{m,t+1}$ (1)	p_m^{outcome} (2)	$PD_{m,t-1,t}$ (3)	$\Delta p_{m,t+1}$ (4)	p_m^{outcome} (5)	$PD_{m,t-1,t}$ (6)
OIB	1.91	22.54		2.95	9.82	
<i>t-statistic</i>	[2.12]	[6.48]		[8.91]	[7.51]	
VOL			−8.79			−2.99
<i>t-statistic</i>			[−6.13]			[−8.41]
Observations	182,873	188,928	182,632	182,873	188,928	182,632
R^2	0.000	0.345	0.009	0.003	0.346	0.010

Notes: This table reports characteristics of the insider accounts and their role in price discovery in the markets they trade. Panel A reports summary statistics for the accounts flagged as insiders, broken out by insider type (timing, conviction, or record) and pooled (*All Insiders*). For each group, the columns report the number of accounts, the share of all markets in which these accounts trade, their share of aggregate profits and aggregate trading volume, the share of accounts that our baseline classification assigns to the skilled, and the share of time bins they make at least one trade in the insiders-traded markets. Panel B reports the price discovery regressions of Sections 3.4 and 3.5, estimated on the markets in which insiders trade, with insiders (columns 1–3) and skilled traders excluding insiders (columns 4–6) as two separate groups. The dependent variables are the next-period price change $\Delta p_{m,t+1}$, the realized outcome p_m^{outcome} , and the change in absolute pricing error $PD_{m,t-1,t}$. OIB is the standardized order imbalance and VOL the standardized trading volume of the respective group. All specifications use lagged price, lagged price change, lagged log volume, and lagged log duration as controls. *t*-statistics are in brackets, with standard errors clustered at the event and month levels.

4.3 Skilled Traders Trade on Public Information

Our last piece of evidence concerns how the skilled trade. An insider profits from private information and has no reason to compete on public information. The skilled, by contrast, compete on public information, and we document three of the strategies they use to do so.

4.3.1 Reacting First to Public News

We start by testing whether skilled traders make profits by reacting quickly to public news. If skilled traders are insiders, their main advantage should come from information that is not yet public. By contrast, if their edge comes from speed, attention, or better information processing, they should trade profitably immediately after the release of public news.

The key empirical challenge is measuring the arrival of news. News arrives at many points throughout a market’s life, and even with fine time bins, it is difficult to isolate the exact moment information reaches the market. We therefore focus on two categories of prediction markets with clearly scheduled information releases: corporate earnings and FOMC announcements.

FOMC meetings occur eight times per year, and corporate earnings are released on pre-scheduled dates; in both cases, there is little pre-announcement leakage, so the release time t can be treated as the moment new information reaches the market. We measure the news itself as the market’s price reaction over a narrow window around the announcement:

$$\text{Surprise}_{m,t} = p_m^{\text{outcome}} - p_{m,t-1}, \quad (9)$$

where $p_{m,t-1}$ is the price just before t and p_m^{outcome} is the final resolution, learned at time t .

We then test whether each group’s order imbalance in the narrow announcement window moves in the direction of the surprise. Let $\Delta\text{OIB}_{m,t}^g$ denote the change in group g ’s order imbalance between the window just before and just after the announcement. We estimate

$$\Delta\text{OIB}_{m,t}^g = \alpha + \beta^g \cdot \text{Surprise}_{m,t} + \varepsilon_{m,t}. \quad (10)$$

A positive β^g means group g shifts toward buying when the surprise is positive and toward

Table 8: Regressions of Order Imbalance Response on News Surprise

Panel A: FOMC Statement Release					
	Skilled (1)	Lucky (2)	Unlucky (3)	Anti-skilled (4)	Market Maker (5)
Surprise	4.99	1.65	−4.65	0.21	−2.19
<i>t-statistic</i>	[3.96]	[0.71]	[−2.31]	[0.30]	[−1.67]
Observations	21	21	21	21	21
R^2	0.25	0.03	0.18	0.00	0.07

Panel B: Corporate Earnings Announcement					
	Skilled (1)	Lucky (2)	Unlucky (3)	Anti-skilled (4)	Market Maker (5)
Surprise	0.17	0.04	−0.01	−0.06	−0.14
<i>t-statistic</i>	[2.62]	[0.63]	[−0.11]	[−1.99]	[−1.99]
Observations	158	158	158	158	158
R^2	0.05	0.00	0.00	0.02	0.04

Notes: This table reports regressions of the change in order imbalance (ΔOIB) of each account group on the news surprise around scheduled announcements, defined as the difference between the final outcome p_m^{outcome} and the price just before the announcement $p_{m,t-1}$. OIB is the net buy volume of the group scaled by total trading volume within a narrow window before and after the news. Panel A covers FOMC statement releases; Panel B covers corporate earnings announcements. Accounts are classified based on the training set; all results are reported on the test set. *t*-statistics are shown in brackets, with standard errors clustered at the FOMC meeting level in Panel A and at the earnings announcement date level in Panel B.

selling when it is negative, indicating that the group is trading in the direction of the news.

Table 8 reports the results. Panel A covers FOMC announcements, and Panel B covers corporate earnings announcements. Only skilled traders' order imbalance responds significantly in the direction of the surprise in both settings. For FOMC meetings, a one percentage point increase in the surprise is associated with a 4.99 percentage point increase in skilled net buying ($t = 3.96$). The magnitude is large because FOMC surprises in these markets are themselves small, with a maximum of about six percentage points. For earnings announcements, a one percentage point increase in the surprise maps to a 17 basis point increase in skilled net buying ($t = 2.62$). Other groups show no consistent response to the surprise, with some coefficients even pointing in the wrong direction.

Appendix E.1 shows that this conclusion holds beyond earnings and FOMC contracts. It exploits Polymarket’s settlement mechanism, which has a dispute window between the time an outcome is first proposed and the time the market is finally settled. Proposed outcomes are rarely overturned, so trading on them during the dispute window is almost always profitable and helps price the contract efficiently. We find that the skilled are the only group that buys the correct outcomes on net. Overall, the edge of the skilled appears to come, in part, from their ability to react efficiently to public news.

4.3.2 Enforcing the Law of One Price

Another strategy available to traders without inside information is arbitrage. Arbitrage trades are valuable because they enforce the law of one price, but they also require capital, monitoring, and execution technology. This makes them a natural source of profits for sophisticated public-information traders, but a less natural avenue for insiders. An insider with valuable private information can trade directly on that information, rather than spend resources capturing small law-of-one-price violations.

Potential arbitrage opportunities arise in multi-outcome events in which exactly one outcome can win. In an event with N mutually exclusive and exhaustive outcomes, a complete basket of Yes tokens pays one USDC at resolution, while a complete basket of No tokens pays $N - 1$ USDC. For No baskets, Polymarket’s conversion mechanism removes the need to wait for resolution, since a trader holding one No share in each of the N outcomes can convert the complete basket into $N - 1$ USDC immediately. The law of one price therefore implies that the No prices should sum to $N - 1$, and if they sum to less, a trader can buy the complete basket and lock in the difference as a riskless profit. To identify these trades in the data, we group each trader’s consecutive trades into blocks, starting a new block whenever the gap between trades exceeds 10 seconds. We classify a block as an executed arbitrage trade if it contains purchases of equal quantities of No tokens across all N outcomes in the same event.

Figure 4 shows that these arbitrage trades are overwhelmingly done by skilled traders. Their net buying volume in arbitrage exceeds 50 million shares, while it is negligible for every other group. That skilled traders expend effort to capture small arbitrage profits is

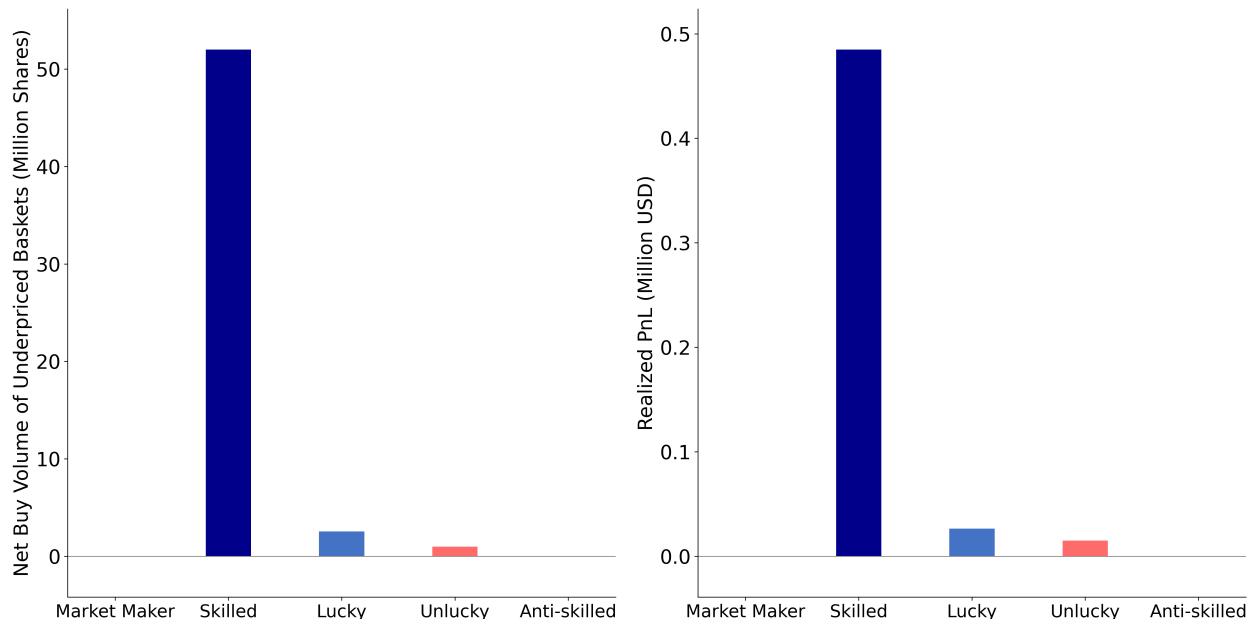


Figure 4: **Arbitrage activity by group.** Panel A reports the net buy volume of underpriced No-token baskets, in millions of shares; Panel B reports the realized PnL from these trades, in millions of USD. Arbitrage opportunities arise in multi-outcome events with N mutually exclusive and exhaustive outcomes, where a complete basket of No tokens is immediately redeemable for $N - 1$ USDC through Polymarket’s conversion mechanism. When the No-token prices sum to less, a trader who buys the complete No basket locks in a riskless profit. We group each trader’s trades into blocks separated by gaps longer than 10 seconds and classify a block as an executed arbitrage trade when it contains purchases of an equal quantity of No tokens across all N outcomes.

hard to square with reliance on inside information, which an insider could trade on directly.

4.3.3 Betting Against Behavioral Mistakes

Finally, a third way skilled traders make money without inside information is by exploiting behavioral biases. One bias documented in many betting markets is the favorite-longshot bias (Griffith, 1949; Thaler and Ziemba, 1988), an empirical pattern in which people like to bet on long shots, where the payoff conditional on winning is high, and avoid favorites, where it is low. As a result, long shots are overpriced and favorites underpriced. Figure E.2 in the Appendix confirms the bias in our sample. For example, contracts priced at 0.1 occur less than 10% of the time, while those at 0.9 occur more than 90%.

To test who exploits this bias, we sort contracts into five price bins, from long shots (prices

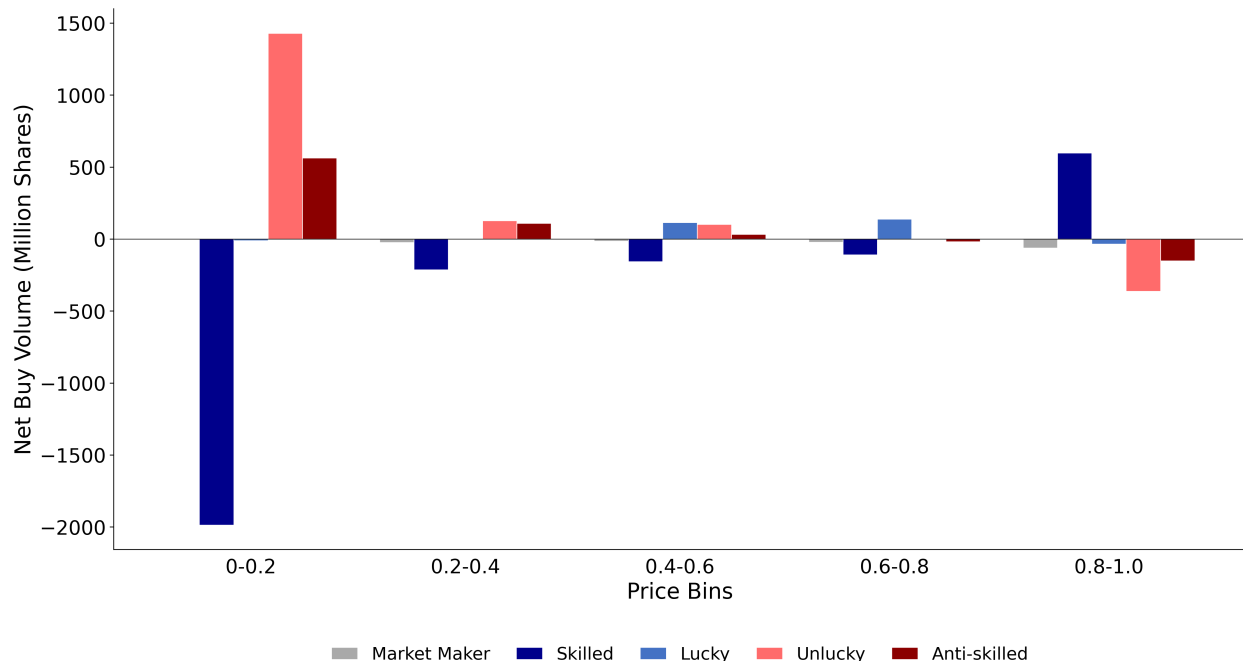


Figure 5: **Net buy volume across the price distribution by group.** Each cluster of bars reports net buy volume (in millions of shares) within a price bin, separately by account group. Contracts are sorted into five bins by price, ranging from long shots (0 to 0.2) to favorites (0.8 to 1). For each bin, the net buy volume is computed for each group. Prices refer to the YES contract; NO trades are normalized to the YES side.

0 to 0.2) to favorites (prices 0.8 to 1), and compute the net buying volume for each group in each bin. All prices are for the YES contract, with NO trades normalized accordingly. Figure 5 shows the result: the skilled bet against the bias, selling overpriced long shots and buying underpriced favorites, mainly at the expense of the unlucky and the anti-skilled.

5 Conclusion

Prediction markets are remarkably accurate, but the source of this accuracy is neither the crowd nor insiders. Using the universe of transactions on Polymarket, we develop a measure of skill and find that only about 3% of accounts are persistently skilled. These traders drive price discovery: their order flow predicts future prices and final outcomes, and their trades move prices toward the correct value. The rest earn profits that are statistically indistinguishable (or worse) from random trading, and those with high realized profits but

no skill tend to reverse into losers out of sample.

These findings are inconsistent with conventional explanations for prediction market accuracy. Accuracy comes not from the crowd or insiders, but from a skilled few who trade on public information by reacting to news as it arrives, enforcing the law of one price, and trading against the crowd's behavioral mistakes. Insiders, who trade on private information, move prices sharply where they appear but are too local to drive market-wide accuracy. The crowd supplies the volume, and the uninformed flow the skilled trade against, but little of the information.

Prediction markets are growing rapidly. As liquidity deepens, event derivatives may serve not only as instruments for speculation, as most trading currently is, but also for hedging event-specific risks that are difficult to manage with conventional financial derivatives. Whether these markets can reliably perform this function depends on the accuracy and integrity of their prices. Our results suggest that this accuracy rests on the continued participation of a small group of skilled traders.

References

- Acharya, V. V. and T. C. Johnson (2007). Insider trading in credit derivatives. *Journal of Financial Economics* 84(1), 110–141.
- Akey, P., V. Grégoire, N. Harvie, and C. Martineau (2026). Who wins and who loses in prediction markets? evidence from polymarket. Working Paper.
- Barber, B. M., X. Huang, T. Odean, and C. Schwarz (2022). Attention-induced trading and returns: Evidence from robinhood users. *Journal of Finance* 77(6), 3141–3190.
- Barber, B. M. and T. Odean (2000). Trading is hazardous to your wealth: The common stock investment performance of individual investors. *The journal of Finance* 55(2), 773–806.
- Barber, B. M., T. Odean, and N. Zhu (2008). Do retail trades move markets? *The Review of Financial Studies* 22(1), 151–186.
- Barras, L., O. Scaillet, and R. Wermers (2010). False discoveries in mutual fund performance: Measuring luck in estimated alphas. *The journal of finance* 65(1), 179–216.
- Barrot, J.-N., R. Kaniel, and D. Sraer (2016). Are retail traders compensated for providing liquidity? *Journal of Financial Economics* 120(1), 146–168.
- Benjamini, Y. and Y. Hochberg (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal statistical society: series B (Methodological)* 57(1), 289–300.
- Berk, J. B. and J. H. Van Binsbergen (2015). Measuring skill in the mutual fund industry. *Journal of financial economics* 118(1), 1–20.
- Boehmer, E., C. M. Jones, X. Zhang, and X. Zhang (2021). Tracking retail investor activity. *The Journal of Finance* 76(5), 2249–2305.
- Bossaerts, F., N. Yadav, P. Bossaerts, C. Nash, T. Todd, T. Rudolf, R. Hutchins, A.-L. Ponsonby, and K. Mattingly (2024). Price formation in field prediction markets: The wisdom in the crowd. *Journal of Financial Markets* 68, 100881.
- Bruggeman, L., A. Katersky, and T. Simpson (2026, May). “survival is at stake”: Inside prediction markets’ race to curb inside trading. <https://abcnews.com/US/inside-race-quell-prediction-markets-insider-trading-woes/story?id=133201587>. ABC News.
- Bürgi, C., W. Deng, and K. Whelan (2025). Makers and takers: The economics of the kalshi prediction market.
- Business Wire (2026, April). Fox to integrate kalshi forecasts across fox news media and fox one platforms. <https://www.morningstar.com/news/business-wire/20260407319986/fox-to-integrate-kalshi-forecasts-across-fox-news-media-and-fox-one-platforms>. Press release.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *The Journal of finance* 52(1), 57–82.
- CBS News (2025, November). Polymarket predictions accuracy: Shayne Coplan on 60 minutes. <https://www.cbsnews.com/news/polymarket-predictions-accuracy-shayne-coplan-60-minutes/>. 60 Minutes interview.

- CNBC (2026, April). Kalshi, polymarket lobby as insider trading, betting eyed by congress. <https://www.cnbc.com/2026/04/15/kalshi-and-polymarket-congress-regulation-washington-influence.html>. CNBC.
- Crypto News Australia (2025, March). Robinhood dives into prediction markets, partners with kalshi to offer political, economic and sports betting. <https://cryptonews.com.au/news/robinhood-dives-into-prediction-markets-partners-with-kalshi-to-offer-political-economic-and-sports-betting-127838/>.
- Della Vedova, J. (2026). Who profits from prediction markets? execution, not information. *Execution, not Information (February 07, 2026)*.
- Diercks, A. M., J. D. Katz, and J. H. Wright (2026). Kalshi and the rise of macro markets. Technical report, National Bureau of Economic Research.
- Eaton, G. W., T. C. Green, B. S. Roseman, and Y. Wu (2022). Retail trader sophistication and stock market quality: Evidence from brokerage outages. *Journal of Financial Economics* 146(2), 502–528.
- Fabi, M., L. Schönleber, V. Ruffo, and R. Marfè (2026). Market efficiency in prediction markets-a comparison with derivatives. *Available at SSRN 6565258*.
- Fama, E. F. and K. R. French (2010). Luck versus skill in the cross-section of mutual fund returns. *The journal of finance* 65(5), 1915–1947.
- Glaze, J., J. López-Vilaró, and A. Stephan (2026). Market expectations of narrative disclosure: Initial evidence from prediction markets. *Available at SSRN 6076807*.
- Glosten, L. R. and P. R. Milgrom (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics* 14(1), 71–100.
- Gomez Cram, R., Y. Guo, T. I. Jensen, and H. Kung (2026). Financial prediction markets: A new measure of earnings expectations. *Available at SSRN 5933475*.
- Griffith, R. M. (1949). Odds adjustments by american horse-race bettors. *The American Journal of Psychology* 62(2), 290–294.
- Grossman, S. J. and J. E. Stiglitz (1980). On the impossibility of informationally efficient markets. *The American economic review* 70(3), 393–408.
- Hartley, J. (2025). Recession prediction markets and rare macroeconomic disaster risk in asset prices. *Available at SSRN 3524686*.
- Harvey, C. R. (2017). Presidential address: The scientific outlook in financial economics. *The Journal of Finance* 72(4), 1399–1440.
- Harvey, C. R., Y. Liu, and H. Zhu (2016). ... and the cross-section of expected returns. *The Review of financial studies* 29(1), 5–68.
- Hvidkjaer, S. (2008). Small trades and the cross-section of stock returns. *The Review of Financial Studies* 21(3), 1123–1151.
- Ivković, Z., C. Sialm, and S. Weisbenner (2008). Portfolio concentration and the performance of individual investors. *Journal of Financial and Quantitative Analysis* 43(3), 613–655.
- Jensen, M. C. (1968). The performance of mutual funds in the period 1945-1964. *The Journal of finance* 23(2), 389–416.

- Kacperczyk, M., C. Sialm, and L. Zheng (2005). On the industry concentration of actively managed equity mutual funds. *Journal of Finance* 60(4), 1983–2011.
- Kacperczyk, M., C. Sialm, and L. Zheng (2008). Unobserved actions of mutual funds. *Review of Financial Studies* 21(6), 2379–2416.
- Kalshi (2023, July). Harnessing the power of prediction markets. <https://news.kalshi.com/p/harnessing-the-power-of-prediction-markets>. Accessed: 2026-04-24.
- Kaniel, R., G. Saar, and S. Titman (2008). Individual investor trading and stock returns. *The Journal of finance* 63(1), 273–310.
- Kelley, E. K. and P. C. Tetlock (2013). How wise are crowds? insights from retail orders and stock returns. *The Journal of Finance* 68(3), 1229–1265.
- Kosowski, R., A. Timmermann, R. Wermers, and H. White (2006). Can mutual fund “stars” really pick stocks? new evidence from a bootstrap analysis. *The Journal of finance* 61(6), 2551–2595.
- Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica: Journal of the Econometric Society*, 1315–1335.
- Lancaster, H. O. (1961). Significance tests in discrete distributions. *Journal of the American Statistical Association* 56(294), 223–234.
- Li, F. and Y. Xie (2025). Trade policy surprises and us convenience yields: Evidence from prediction markets. *Available at SSRN 5882222*.
- Manski, C. F. (2006). Interpreting the predictions of prediction markets. *economics letters* 91(3), 425–429.
- Mitts, J. and M. Ofir (2026). From iran to taylor swift: Informed trading in prediction markets. *Available at SSRN*.
- Ng, H., L. Peng, Y. Tao, and D. Zhou (2025). Price discovery and trading in prediction markets. *Available at SSRN 5331995*.
- Polymarket (2026). Polymarket vs. polling. <https://help.polymarket.com/en/articles/13364238-polymarket-vs-polling>. Accessed: 2026-04-24.
- Reichenbach, F. and M. Walther (2025). Exploring decentralized prediction markets: Accuracy, skill, and bias on polymarket. Working Paper.
- Routledge, R. (1994). Practicing safe statistics with the mid-p. *Canadian Journal of Statistics* 22(1), 103–110.
- Sirolly, A., H. Ma, Y. Kanoria, and R. Sethi (2025). Network-based detection of wash trading. *Available at SSRN*.
- Snowberg, E., J. Wolfers, and E. Zitzewitz (2011). How prediction markets can save event studies. *Working Paper, University of Utah*.
- Surowiecki, J. (2004). The wisdom of crowds: Why the many are smarter than the few and how collective wisdom shapes business. *Economies, Societies and Nations* 296(5), 1–27.
- Swanson, E. T., R. Wang, and Y. Wu (2025). The effects of monetary policy on macroeconomic expectations: High-frequency evidence from traded event contracts. Technical report, Working Paper.

- Thaler, R. H. and W. T. Ziemba (1988). Anomalies: Parimutuel betting markets: Racetracks and lotteries. *Journal of Economic perspectives* 2(2), 161–174.
- Tsang, K. P. and Z. Yang (2026). The anatomy of polymarket: Evidence from the 2024 presidential election. *arXiv preprint arXiv:2603.03136*.
- Welch, I. (2022). The wisdom of the robinhood crowd. *Journal of Finance* 77(3), 1489–1527.
- Wertheim, J., A. Chasan, A. Bast, and J. Kegu (2026, May). Suspected insider accounts net \$2.4 million on Polymarket Iran war bets with 98% win rate. <https://www.cbsnews.com/news/betting-on-iran-war-insider-trading-concerns-prediction-markets-60-minutes/>. 60 Minutes.
- Wolfers, J. and E. Zitzewitz (2004). Prediction markets. *Journal of economic perspectives* 18(2), 107–126.
- Wolfers, J. and E. Zitzewitz (2006). Interpreting prediction market prices as probabilities.
- Yang, H.-C. A. (2026). Skilled liquidity provision in prediction markets: Evidence from 150 million trades. *Available at SSRN 6396698*.

Online Appendix

Prediction Market Accuracy: Crowd Wisdom or Informed Minority?

Roberto Gómez-Cram Yunhan Guo Theis Ingerslev Jensen Howard Kung

Appendix Table of Contents

- A. [Model: Crowd Wisdom or Informed Minority?](#)
- B. [Robustness of the Skill Measure](#)
- C. [Insider Trading](#)
- D. [Mutual Fund Skill Persistence](#)
- E. [Additional Analysis](#)

A Model: Crowd Wisdom or Informed Minority?

We develop a simple model, building on [Kyle \(1985\)](#), to distinguish between a world in which prediction market accuracy comes from the crowd and one in which it comes from an informed minority. Prediction-market platforms themselves often describe their accuracy in wisdom-of-crowds terms. Kalshi writes that prediction markets “leverage the wisdom of crowds” by aggregating the “knowledge, information, and insights of a diverse group of participants” ([Kalshi, 2023](#)). Similarly, Polymarket states that prediction markets can be more accurate than polls because of the “collective intelligence” of diverse participants ([Polymarket, 2026](#)). This view echoes the classic Francis Galton ox-weight example emphasized by [Surowiecki \(2004\)](#), in which the average guess of a large crowd came strikingly close to the truth. Motivated by this view, we model the wisdom of crowds as a benchmark in which all traders receive independent signals that are ex-ante equally informative. We contrast this benchmark with the more conventional explanation of price informativeness in financial markets, in which a subset of informed traders move prices toward the truth and earn profits at the expense of uninformed traders.

We consider a one-period asset with payoff $v \sim \mathcal{N}(\bar{v}, \sigma_v^2)$, traded by N risk-neutral traders and a competitive risk-neutral market maker. We assume normality to stay close to [Kyle \(1985\)](#) and to obtain a tractable linear equilibrium. The economic distinction between the two cases below comes from who is informed rather than the Gaussian assumption. Traders are divided into two groups: the first $K \leq N$ are informed, and the remaining $N - K$ are uninformed. Each informed trader $i \in \{1, \dots, K\}$ privately observes

$$s_i = v + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \tau^{-1}),$$

where $\tau > 0$ is the common signal precision. The signal errors are independent across traders and independent of v . Informed traders choose a trading intensity β_i strategically to maximize $\mathbb{E}[\pi_i \mid s_i]$, so their demand is

$$x_i = \beta_i(s_i - \bar{v}) + \eta_i, \quad \eta_i \sim \mathcal{N}(0, \sigma_\eta^2),$$

where $\beta_i(s_i - \bar{v})$ is the signal-based component and η_i is an exogenous non-informational component capturing motives such as liquidity demand, hedging needs, or behavioral trading. Uninformed traders $i \in \{K+1, \dots, N\}$ observe no signal and trade only for non-informational reasons, so $x_i = \eta_i$. Each trader's realized profit is $\pi_i = x_i(v - p)$.

The market maker observes only total order flow, $y = \sum_{i=1}^N x_i$, but cannot distinguish between the informational and non-informational components. Competition and risk neutrality imply that the market maker has zero expected profit, so the price satisfies $p = E[v \mid y]$.

We focus on two special cases:

Assumption 1 (Wisdom of Crowds). Every trader is informed: $K = N$.

Every trader observes an independent private signal and trades strategically on it. The price, therefore, aggregates N dispersed pieces of information about v .

Assumption 2 (Wisdom of the Informed). A subset of traders is informed: $1 \leq K < N$.

Only a subset of traders observes informative signals, while the remaining $N - K$ traders trade for non-informational reasons. The price, therefore, reflects the information held by the informed subset.⁸

Under either information structure, the market price is informative in the sense that $\text{var}(v \mid p) < \sigma_v^2$. Both theories can therefore explain why prediction markets produce accurate prices. They differ, however, in their predictions about who earns profits, whether those profits persist, and who contributes to price informativeness (proofs are in Appendix A.1):

Proposition 1 (Distinguishing Crowds from the Informed). *In the linear equilibrium:*

(a) Cross-section of profits. *Under the wisdom of crowds (Assumption 1),*

$$E[\pi_i] = 0 \quad \text{for all } i.$$

Under the wisdom of the informed (Assumption 2),

$$E[\pi_i] > 0 \quad \text{for informed traders } (i \leq K) \quad \text{and} \quad E[\pi_j] < 0 \quad \text{for uninformed traders } (j > K).$$

⁸Informed traders play the role of the insider in Kyle (1985), while uninformed traders play the role of noise traders. Our setup differs from Kyle (1985) in two ways: we allow multiple informed traders with noisy signals rather than a single monopolistic insider, and we give each trader an individual η_i rather than modeling noise trading as a single aggregate order.

(b) Profit persistence. Consider a sequence of independent markets with independent realizations $(v_m, \varepsilon_{i,m}, \eta_{i,m})$, and let $\mu_i \equiv \mathbb{E}[\pi_{i,m} \mid \text{type of } i]$ denote trader i 's expected profit per market given type. The population slope in a cross-sectional regression of future average profit on past average profit is

$$b = \frac{\text{var}_i(\mu_i)}{\text{var}_i(\mu_i) + \bar{\sigma}_\pi^2 / M_{\text{past}}}, \quad (\text{A.1})$$

where $\bar{\sigma}_\pi^2 \equiv \mathbb{E}_i[\text{var}(\pi_{i,m} \mid \text{type of } i)]$ is the average within-type variance of per-market profits.

(c) Concentration of price discovery. Measured by each trader's contribution to $\text{cov}(v, p)$, every trader contributes equally under the wisdom of crowds. Under the wisdom of the informed, only the K informed traders contribute. If no trader has private information, then total order flow is purely non-informational, $y = u = \sum_{j=1}^N \eta_j$, which is independent of v , and therefore $\text{var}(v \mid p) = \sigma_v^2$.

Part (a) captures the idea that informed traders earn profits only when they trade against less-informed counterparties. Under the wisdom of crowds, every trader is equally informed, so no trader has an informational advantage, and expected profits are zero. Under the wisdom of the informed, informed traders profit at the expense of uninformed traders. Part (b) then shows that cross-sectional differences in realized profits are pure luck under the wisdom of crowds, but reflect persistent differences in expected profits under the wisdom of the informed. Part (c) shows that price informativeness is generated symmetrically across all traders under the wisdom of crowds, whereas under the wisdom of the informed, it is generated entirely by the informed subset. The paper takes these predictions to the data, showing strong evidence that prediction market accuracy is driven by a small minority of traders, which is inconsistent with assumption 1 but consistent with assumption 2.

Next, we show that informed traders' order imbalance conveys information about the future outcome beyond what is incorporated into prices under the wisdom of the informed, but not under the wisdom of crowds. To do so, we define informed traders' total order imbalance as

$$w \equiv \sum_{i=1}^K x_i = B(v - \bar{v}) + \beta \sum_{i=1}^K \varepsilon_i + \sum_{i=1}^K \eta_i,$$

where $B \equiv \sum_{i=1}^K \beta_i = K\beta$. Total order flow decomposes as $y = w + (y - w)$, where $y - w = \sum_{i=K+1}^N \eta_i$ is the uninformed traders' order flow. Because the η_i are independent across traders and of v , the term $y - w$ is independent of both v and w . This setup implies the following result:

Proposition 2 (Order-flow informativeness beyond price). *In the linear equilibrium:*

(a) Wisdom of crowds (Assumption 1, $K = N$). *Every trader is informed, so $w = y$ and the price is a sufficient statistic for the outcome:*

$$E[v \mid p, w] = E[v \mid p] = p, \quad \text{var}(v \mid p, w) = \text{var}(v \mid p).$$

The informed order imbalance contains no information about the outcome beyond the price.

(b) Wisdom of the informed (Assumption 2, $K < N$). *The informed order imbalance both shifts and sharpens the forecast:*

$$E[v \mid p, w] \neq p, \quad \text{var}(v \mid p, w) < \text{var}(v \mid p).$$

Stated differently, in the population linear projection of the outcome v on p and w , the coefficient on w is strictly positive. The informed order imbalance is therefore incrementally informative about the outcome beyond the price.

The intuition for Proposition 2 is that the market maker pools the informed orders w with the uninformed orders $y - w$ and cannot tell them apart, so prices do not fully reflect w . An observer who also sees the informed order imbalance w strips out the uninformed noise and forms a better forecast of the outcome than the price alone can provide. Under the wisdom of crowds, there are no uninformed traders, so $w = y$, meaning that the price already reflects the informed order flow, and there is nothing to extract.

To understand the implications of the proposition, note that a conditional variance is the expected squared error of the best forecast, $\text{var}(v \mid X) = E[(v - E[v \mid X])^2]$. Therefore, part (a) says that adding the informed order imbalance to the price leaves the forecast error for the outcome unchanged, whereas part (b) says it strictly lowers it. This is what our price-discovery tests estimate. Regressing the realized outcome on a group's order imbalance while

controlling for the price, the strict inequality $\text{var}(v \mid p, w) < \text{var}(v \mid p)$ is equivalent to a non-zero coefficient on the informed group's order imbalance, which we show to be positive. A positive and significant coefficient is therefore empirical evidence against assumption 1 but for assumption 2.

A.1 Proofs

A.1.1 Equilibrium Derivation

Traders submit market orders in a single trading round, and the market maker sets the price as a function of aggregate order flow. We solve for a linear equilibrium.

Equilibrium concept. A linear equilibrium consists of trading intensities $\{\beta_i\}_{i=1}^K$ for informed traders and a pricing rule such that: (i) each informed trader chooses β_i to maximize expected profit conditional on her signal, $E[\pi_i \mid s_i]$, taking the other traders' strategies and the pricing rule as given; and (ii) the pricing rule satisfies $p = E[v \mid y]$. As we show below, joint normality of (v, y) implies that the equilibrium pricing rule is linear in order flow, so $p = \bar{v} + \lambda y$.

Trader optimization. Informed traders choose β_i to maximize $E[\pi_i \mid s_i]$. Write the pricing error as

$$v - p = (v - \bar{v})(1 - \lambda B) - \lambda \sum_{i=1}^K \beta_i \varepsilon_i - \lambda u,$$

where $B \equiv \sum_{i=1}^K \beta_i$ and $u \equiv \sum_{i=1}^N \eta_i$ is aggregate non-informational order flow (formalized in the next paragraph). Define the Bayesian posterior weight $\delta \equiv \tau/(\tau_0 + \tau)$, where $\tau_0 = 1/\sigma_v^2$ is the prior precision. Then $E[v - \bar{v} \mid s_i] = \delta(s_i - \bar{v})$. Using also $E[\varepsilon_i \mid s_i] = (1 - \delta)(s_i - \bar{v})$ and $E[\varepsilon_j \mid s_i] = 0$ for $j \neq i$, we obtain

$$E[v - p \mid s_i] = \left[\delta - \lambda \delta \sum_{j \neq i} \beta_j - \lambda \beta_i \right] (s_i - \bar{v}).$$

The non-informational component contributes the term $E[\eta_i(v - p) \mid s_i] = -\lambda\sigma_\eta^2$ to expected profit. Because this term does not depend on β_i , it drops out of the first-order condition. Maximizing $E[\pi_i \mid s_i]$ with respect to β_i and imposing the symmetric fixed point $\beta_i = \beta$ yields

$$\beta = \frac{\delta(1 - \lambda(K - 1)\beta)}{2\lambda}. \quad (\text{A.2})$$

Uninformed traders observe no signal, so their demand reduces to $x_j = \eta_j$ (equivalently, $\beta_j = 0$).

Market maker. Aggregating demands across all N traders, total order flow is

$$y = B(v - \bar{v}) + \sum_{i=1}^K \beta_i \varepsilon_i + u,$$

where $u = \sum_{i=1}^N \eta_i$. Since the η_i are mutually independent, $\text{var}(u) = N\sigma_\eta^2$. Because all random variables are jointly normal and $E[y] = 0$, the conditional expectation $E[v \mid y]$ is linear in y :

$$\lambda = \frac{\text{cov}(v, y)}{\text{var}(y)} = \frac{B\sigma_v^2}{B^2\sigma_v^2 + \sum_{i=1}^K \beta_i^2/\tau + N\sigma_\eta^2}. \quad (\text{A.3})$$

Because informed traders share the same signal precision τ , symmetry implies $\beta_i = \beta$ for all informed traders. Substituting this into the expression above gives

$$\lambda = \frac{K\beta\sigma_v^2}{K^2\beta^2\sigma_v^2 + K\beta^2/\tau + N\sigma_\eta^2}. \quad (\text{A.4})$$

Solution under Assumption 1. When $K = N$, Equation (A.2) implies $\lambda\beta = \delta/(2 + (N - 1)\delta)$. Substituting this relation into Equation (A.4) and solving yields the closed-form equilibrium

$$\beta = \frac{\sigma_\eta\sqrt{\delta}}{\sigma_v}, \quad \lambda = \frac{\sigma_v\sqrt{\delta}}{(2 + (N - 1)\delta)\sigma_\eta}. \quad (\text{A.5})$$

Solution under Assumption 2. When $1 \leq K < N$, only the K informed traders contribute signal-based order flow, whereas all N traders contribute non-informational order flow. Solving yields

$$\beta = \frac{\sigma_\eta \sqrt{\delta N/K}}{\sigma_v}, \quad \lambda = \frac{\sigma_v \sqrt{\delta K/N}}{(2 + (K-1)\delta) \sigma_\eta}. \quad (\text{A.6})$$

A.1.2 Proof of Proposition 1

The proof uses the equilibrium objects (β, λ) derived above.

Part (a). Decompose each trader's expected profit into a signal-based component and a non-informational component. For informed trader i ,

$$\mathbb{E}[\pi_i] = \underbrace{\beta \mathbb{E}[(s_i - \bar{v})(v - p)]}_{\text{signal-based gain}} + \underbrace{\mathbb{E}[\eta_i(v - p)]}_{\text{noise loss}},$$

while uninformed traders have only the second term. The non-informational component is a loss of $-\lambda\sigma_\eta^2$ for every trader. The reason is that η_i is independent of v but correlated with p through total order flow y , and therefore $\mathbb{E}[\eta_i p] = \lambda \mathbb{E}[\eta_i y] = \lambda\sigma_\eta^2$.

The competitive market maker earns zero expected profit. Since $p = \mathbb{E}[v \mid y]$, we have $\mathbb{E}[v - p \mid y] = 0$, and therefore

$$\sum_{i=1}^N \mathbb{E}[\pi_i] = \mathbb{E}[y(v - p)] = \mathbb{E}[y \cdot \mathbb{E}[v - p \mid y]] = 0.$$

Uninformed traders have only the non-informational loss, so $\mathbb{E}[\pi_j] = -\lambda\sigma_\eta^2$ for $j > K$. By symmetry across informed traders, combining this with the zero-sum identity yields

$$\mathbb{E}[\pi_i] = \frac{N - K}{K} \lambda \sigma_\eta^2$$

for each informed trader. This quantity is strictly positive under Assumption 2 and equal to zero under Assumption 1 ($K = N$), where symmetry implies zero expected profit for every trader.

Part (b). Because markets are independent across m and equilibrium outcomes depend on a trader only through her type, $\mu_i = \mathbb{E}[\pi_{i,m} \mid \text{type of } i]$ is constant across markets for a given trader. Under Assumption 1, all traders have the same type, so Part (a) implies $\mu_i = 0$ for every i and hence $\text{var}_i(\mu_i) = 0$. Under Assumption 2, μ_i takes two distinct values, $\mu_H > 0$ for informed traders and $\mu_U < 0$ for uninformed traders, by Part (a), so $\text{var}_i(\mu_i) > 0$.

To derive the slope in Equation (A.1), write $\pi_{i,m} = \mu_i + \xi_{i,m}$, where $\xi_{i,m}$ is a mean-zero residual that captures randomness from v_m , $\varepsilon_{i,m}$, and $\eta_{i,m}$. These residuals are independent across markets and satisfy $\mathbb{E}[\xi_{i,m} \mid \text{type of } i] = 0$. Let $\bar{\pi}_i^{\text{past}}$ and $\bar{\pi}_i^{\text{future}}$ denote trader i 's average profit across two disjoint sets of markets of sizes M_{past} and M_{future} . Then the cross-sectional covariance is $\text{cov}_i(\bar{\pi}_i^{\text{past}}, \bar{\pi}_i^{\text{future}}) = \text{var}_i(\mu_i)$, while the cross-sectional variance of past average profits is $\text{var}_i(\mu_i) + \bar{\sigma}_\pi^2/M_{\text{past}}$. The population slope in a cross-sectional regression of future average profit on past average profit is therefore their ratio, which yields Equation (A.1). Hence $b = 0$ under Assumption 1, whereas $b > 0$ under Assumption 2. As $M_{\text{past}} \rightarrow \infty$, the attenuation term $\bar{\sigma}_\pi^2/M_{\text{past}}$ vanishes, so $b \rightarrow 1$.

Part (c). The signal-based component of order flow is $\beta \sum_{i=1}^K (s_i - \bar{v})$. Under Assumption 1 ($K = N$), $\text{cov}(v, p) = \lambda \text{cov}(v, y) = \lambda K \beta \sigma_v^2 = N \lambda \beta \sigma_v^2$, so each trader contributes equally, namely $\lambda \beta \sigma_v^2 = \text{cov}(v, p)/N$. Under Assumption 2, only the K informed traders have nonzero trading intensity. If no trader has private information, then total order flow is purely non-informational, $y = u = \sum_{j=1}^N \eta_j$, which is independent of v . Hence $p = \mathbb{E}[v \mid y] = \bar{v}$ and $\text{var}(v \mid p) = \sigma_v^2$. $\square$

A.1.3 Proof of Proposition 2

The proof uses the equilibrium objects (β, λ) and the fact that $\beta > 0$.

Covariances. Because the signal errors ε_i and the non-informational components η_i are independent of v ,

$$\text{cov}(v, w) = B \sigma_v^2 = \text{cov}(v, y).$$

Adding the uninformed orders $y - w$ to w leaves the covariance with v unchanged, since $y - w$ is independent of v .

Variances. Using $\text{var}(\varepsilon_i) = \tau^{-1}$ and $\text{var}(\eta_i) = \sigma_\eta^2$, and mutual independence,

$$\text{var}(w) = B^2 \sigma_v^2 + K \beta^2 / \tau + K \sigma_\eta^2, \quad \text{var}(y) = \text{var}(w) + (N - K) \sigma_\eta^2.$$

Hence $\text{var}(w) \leq \text{var}(y)$, with strict inequality if and only if $K < N$.

Conditioning on (p, w) . Since $p = \bar{v} + \lambda y$ with $\lambda \neq 0$, the pair (p, w) carries the same information as $(w, y - w)$. The component $y - w = \sum_{i=K+1}^N \eta_i$ is independent of (v, w) , so it is uninformative about v once w is known. Therefore

$$\mathbb{E}[v \mid p, w] = \mathbb{E}[v \mid w], \quad \text{var}(v \mid p, w) = \text{var}(v \mid w).$$

Comparison. All variables are jointly normal, so the conditional variances are

$$\text{var}(v \mid w) = \sigma_v^2 - \frac{(B \sigma_v^2)^2}{\text{var}(w)}, \quad \text{var}(v \mid p) = \text{var}(v \mid y) = \sigma_v^2 - \frac{(B \sigma_v^2)^2}{\text{var}(y)},$$

where the second equality uses that p is an invertible affine function of y . Since $\text{cov}(v, w) = \text{cov}(v, y) = B \sigma_v^2$ and $\text{var}(w) \leq \text{var}(y)$, the subtracted term is larger for w , so

$$\text{var}(v \mid p, w) = \text{var}(v \mid w) \leq \text{var}(v \mid p),$$

with strict inequality if and only if $K < N$. This is Part (b). When $K = N$ (Assumption 1), $w = y$, so $\text{var}(w) = \text{var}(y)$ and the inequality holds with equality; moreover $\mathbb{E}[v \mid p, w] = \mathbb{E}[v \mid w] = \mathbb{E}[v \mid y] = p$, which is Part (a).

Projection coefficient. For the equivalent statement in Part (b), normality gives

$$\mathbb{E}[v \mid p, w] = \mathbb{E}[v \mid w] = \bar{v} + \frac{B \sigma_v^2}{\text{var}(w)} w,$$

so the population linear projection of v on (p, w) loads $B\sigma_v^2/\text{var}(w) > 0$ on w .⁹ □

B Robustness of the Skill Measure

In this appendix we show that our main results are robust to alternative ways of measuring skill and of identifying market makers. We re-run the price-discovery tests under each of the resulting classifications. Section B.1 varies the event filter and the multiple-testing adjustment. Section B.2 replaces the sign-randomization null with alternative benchmarks for an uninformed trader. Section B.3 varies the criteria that identify market makers. In every specification, the skilled group’s order flow predicts price changes and final outcomes and reduces pricing errors, while the other groups’ does not.

B.1 Event Filter and Multiple Testing

In the main text, we classify an account as skilled if it trades in at least ten events and has a sign-randomization p -value below 0.05, with no adjustment for multiple testing. We now vary the two classification choices behind this definition, the event filter and the multiple-testing adjustment. Table B.1 reports the price-discovery results under three combinations of these choices. Panel A keeps the ten-event filter but adds the Benjamini–Hochberg adjustment; Panel B drops the filter and uses no adjustment; Panel C drops the filter and adds the adjustment. In every panel, the skilled group’s order flow predicts next-period price changes and final outcomes and reduces pricing errors, while the other groups’ does not.

⁹In this one-period model, the projection puts zero weight on p , because once w is known, the price is w plus independent noise. In a dynamic market, the price would likely enter with a non-zero weight, because it contains information about past informed order flow.

Table B.1: Event Filter and Multiple-Testing Robustness

Panel A: Ten-event Filter, Benjamini–Hochberg Adjustment						
	Skilled		$PD_{m,t-1,t}$			
	$\Delta p_{m,t+1}$ (1)	p_m^{outcome} (2)	Skilled (3)	Lucky (4)	Unlucky (5)	Anti-skilled (6)
OIB	2.39	9.07				
<i>t-statistic</i>	[10.64]	[7.38]				
VOL			−6.61	0.94	2.59	−0.72
<i>t-statistic</i>			[−4.88]	[1.96]	[6.37]	[−0.88]
Observations	1,267,183	1,362,338	1,265,693	1,265,693	1,265,693	1,265,693
R^2	0.001	0.326	0.032	0.023	0.026	0.022
Panel B: No Event Filter, No Adjustment						
	Skilled		$PD_{m,t-1,t}$			
	$\Delta p_{m,t+1}$ (1)	p_m^{outcome} (2)	Skilled (3)	Lucky (4)	Unlucky (5)	Anti-skilled (6)
OIB	2.03	7.78				
<i>t-statistic</i>	[13.03]	[9.62]				
VOL			−4.93	1.65	2.48	0.79
<i>t-statistic</i>			[−5.53]	[2.97]	[6.39]	[1.74]
Observations	1,267,183	1,362,338	1,265,693	1,265,693	1,265,693	1,265,693
R^2	0.001	0.327	0.031	0.024	0.025	0.022
Panel C: No Event Filter, Benjamini–Hochberg Adjustment						
	Skilled		$PD_{m,t-1,t}$			
	$\Delta p_{m,t+1}$ (1)	p_m^{outcome} (2)	Skilled (3)	Lucky (4)	Unlucky (5)	Anti-skilled (6)
OIB	2.33	9.99				
<i>t-statistic</i>	[10.70]	[6.68]				
VOL			−6.53	0.67	2.60	−1.40
<i>t-statistic</i>			[−4.97]	[1.60]	[6.40]	[−1.39]
Observations	1,267,183	1,362,338	1,265,693	1,265,693	1,265,693	1,265,693
R^2	0.001	0.327	0.031	0.023	0.026	0.022

Notes: This table reports the price-discovery regressions under alternative event filters and multiple-testing corrections. Each column is a separate regression estimated on the test set, with accounts classified on the training set. Columns (1)–(2) regress the next-period price change $\Delta p_{m,t+1}$ and the final outcome p_m^{outcome} on the skilled group’s order imbalance (OIB); columns (3)–(6) regress the price-discovery contribution $PD_{m,t-1,t}$, the change in a market’s absolute pricing error defined in Section 3.5, on each group’s volume share (VOL). Each panel reclassifies accounts under the event filter and multiple-testing rule named in its heading. Controls are lagged price, lagged price change, lagged log volume, and log market duration. *t*-statistics are in brackets, with standard errors clustered by event and month.

B.2 Alternative Benchmarks for an Uninformed Trader

In the main text, we measure skill against a sign-randomization null. The null treats an uninformed trader as one who picks the direction of each trade at random while holding the size of each trade and the market’s outcome fixed. A trader is skilled if their realized PnL is too large to arise under this null.

One concern is that the sign-randomization null fixes the size of each trade, so a trader could be classified as skilled for how much they staked rather than for which side they took. We therefore randomize size along with direction. We flip each trade’s direction with probability one half and reshuffle trade sizes, both at the event level, and cap any single trade at ten percent of the event’s volume, so the uninformed benchmark now picks both side and stake at random. Panel A of Table B.2 shows the results. Under this classification, the skilled still predict future prices and final outcomes and their trades reduce pricing errors, while the other groups’ trades instead increase them, just as under the benchmark classification.

Another concern is that the null fixes the market’s outcome, so a trader who only backs heavy favorites can be classified as skilled without any edge. For example, an account that buys 0.99 favorites wins almost every bet, so over a short record it most likely wins them all, even though its rare losses would surface over a longer one. We address this by randomizing the outcome rather than the trade, so that correctly backing a favorite no longer counts as skill.

We hold each trade fixed in size, direction, and timing, and resample the market outcomes, drawing each outcome from a Bernoulli distribution with the trade’s execution price as the success probability. Panel B of Table B.2 shows the results. Under this classification, the skilled still predict future prices and final outcomes and their trades reduce pricing errors, while the other groups’ trades instead increase them, just as under the benchmark classification.

Table B.2: Alternative Uninformed-Trader Benchmarks

Panel A: Random Trade Direction and Size						
	Skilled		$PD_{m,t-1,t}$			
	$\Delta p_{m,t+1}$ (1)	p_m^{outcome} (2)	Skilled (3)	Lucky (4)	Unlucky (5)	Anti-skilled (6)
OIB	1.56	7.71				
<i>t-statistic</i>	[16.04]	[5.83]				
VOL			−6.67	0.71	2.51	0.66
<i>t-statistic</i>			[−10.21]	[2.61]	[6.13]	[1.92]
Observations	1,267,183	1,362,338	1,265,693	1,265,693	1,265,693	1,265,693
R^2	0.000	0.326	0.032	0.023	0.025	0.022
Panel B: Resampled Market Outcomes						
	Skilled		$PD_{m,t-1,t}$			
	$\Delta p_{m,t+1}$ (1)	p_m^{outcome} (2)	Skilled (3)	Lucky (4)	Unlucky (5)	Anti-skilled (6)
OIB	2.16	5.04				
<i>t-statistic</i>	[12.69]	[4.25]				
VOL			−5.88	1.84	2.66	0.10
<i>t-statistic</i>			[−9.98]	[4.38]	[6.15]	[0.31]
Observations	1,267,183	1,362,338	1,265,693	1,265,693	1,265,693	1,265,693
R^2	0.001	0.326	0.034	0.024	0.026	0.022

Notes: This table reports the price-discovery regression results under alternative definitions of the uninformed-trader benchmark used to classify accounts: random trade direction and size (Panel A) and resampled market outcomes (Panel B). Each column is a separate regression estimated on the test set, with accounts classified on the training set. Columns (1)–(2) regress the next-period price change $\Delta p_{m,t+1}$ and the final outcome p_m^{outcome} on the skilled group's order imbalance (OIB); columns (3)–(6) regress the price-discovery contribution $PD_{m,t-1,t}$, the change in a market's absolute pricing error defined in Section 3.5, on each group's volume share (VOL). Each panel reclassifies accounts under the uninformed-trader benchmark named in its heading. Controls are lagged price, lagged price change, lagged log volume, and log market duration. *t*-statistics are in brackets, with standard errors clustered by event and month.

B.3 Market-Maker Criteria

In the main text, we identify market makers by their behavior rather than by the skill test. An account is a market maker if it trades in at least 100 markets, draws at least 70% of its volume from limit orders, and trades both sides of an outcome in at least 70% of the markets it trades.

We vary each of these three thresholds, raising and lowering the market-count threshold to 500 and to 10 and the limit-order and both-sides thresholds to 80% and to 50%. Table B.3 reports how many accounts each definition assigns to the market-maker and skilled groups. The market-maker count is sensitive to the thresholds, ranging from about 600 accounts when we require 500 markets to 7,700 when we require only 10. The skilled count barely moves, holding near 54,000 accounts and about 3.1% of all accounts throughout. Because market makers are a relatively small group, redrawing its boundary reshuffles few accounts and leaves the skilled essentially untouched.

Table B.4 re-runs the price-discovery tests for the two groups under each definition. The skilled drive price discovery throughout, with their order imbalance predicting both next-period price changes and final outcomes and their volume reducing pricing errors. Market makers' order imbalance also predicts next-period price changes, but it predicts final outcomes with the opposite sign, and their volume does not reduce pricing errors. Their order flow tracks short-run price moves without anticipating the final outcome or improving accuracy, consistent with their role as liquidity providers rather than informed traders.

Table B.3: Account Counts under Alternative Market-Maker Definitions

	Market-count threshold				Order thresholds			
	≥ 500		≥ 10		$\geq 80\%$		$\geq 50\%$	
	Accounts	Share	Accounts	Share	Accounts	Share	Accounts	Share
	(000s)	(%)	(000s)	(%)	(000s)	(%)	(000s)	(%)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Market Maker	0.62	0.04	7.70	0.45	0.84	0.05	4.67	0.27
Skilled	54.50	3.17	53.58	3.11	54.49	3.16	53.87	3.13

Notes: This table reports the number and share of market makers and skilled under alternative market-maker definitions. Each definition changes one criterion from the main-text definition, which requires at least 100 markets, 70% limit-order volume, and both-sides trading in at least 70% of markets. The first two columns vary the market-count threshold; the last two vary the limit-order and both-sides thresholds jointly, holding the others at their main-text values. **Accounts** is the number of unique accounts in the group, in thousands; **Share** is the percentage of all accounts.

Table B.4: Alternative Market-Maker Definitions

	Skilled			Market Makers		
	$\Delta p_{m,t+1}$ (1)	p_m^{outcome} (2)	$PD_{m,t-1,t}$ (3)	$\Delta p_{m,t+1}$ (4)	p_m^{outcome} (5)	$PD_{m,t-1,t}$ (6)
Panel A: ≥ 500 Markets, $\geq 70\%$ Limit Orders, $\geq 70\%$ Both Sides						
OIB	2.09	7.31		1.21	-5.95	
<i>t-statistic</i>	[12.99]	[9.34]		[11.87]	[-10.91]	
VOL			-4.64			-1.23
<i>t-statistic</i>			[-5.19]			[-1.89]
Observations	1,267,183	1,362,338	1,265,693	1,267,183	1,362,338	1,265,693
R^2	0.002	0.327	0.031	0.000	0.326	0.023
Panel B: ≥ 10 Markets, $\geq 70\%$ Limit Orders, $\geq 70\%$ Both Sides						
OIB	2.04	7.78		1.23	-5.64	
<i>t-statistic</i>	[12.96]	[9.53]		[12.11]	[-10.55]	
VOL			-4.97			-0.56
<i>t-statistic</i>			[-5.56]			[-1.21]
Observations	1,267,183	1,362,338	1,265,693	1,267,183	1,362,338	1,265,693
R^2	0.001	0.327	0.032	0.001	0.326	0.022
Panel C: ≥ 100 Markets, $\geq 80\%$ Limit Orders, $\geq 80\%$ Both Sides						
OIB	2.65	6.78		0.98	-5.33	
<i>t-statistic</i>	[21.96]	[14.46]		[7.81]	[-11.10]	
VOL			-5.01			0.10
<i>t-statistic</i>			[-5.57]			[0.18]
Observations	1,267,183	1,362,338	1,265,693	1,267,183	1,362,338	1,265,693
R^2	0.003	0.327	0.034	0.000	0.326	0.022
Panel D: ≥ 100 Markets, $\geq 50\%$ Limit Orders, $\geq 50\%$ Both Sides						
OIB	1.26	10.15		1.89	-3.09	
<i>t-statistic</i>	[9.35]	[8.82]		[23.90]	[-7.73]	
VOL			-6.03			-0.40
<i>t-statistic</i>			[-6.44]			[-1.22]
Observations	1,267,183	1,362,338	1,265,693	1,267,183	1,362,338	1,265,693
R^2	0.000	0.327	0.032	0.002	0.326	0.022

Notes: This table reports the price-discovery regression results under alternative market-maker definitions. Each column is a separate regression estimated on the test set, with accounts classified on the training set. For each group, the first two columns regress the next-period price change $\Delta p_{m,t+1}$ and the final outcome p_m^{outcome} on the group's order imbalance (OIB), and the third regresses the price-discovery contribution $PD_{m,t-1,t}$, the change in a market's absolute pricing error defined in Section 3.5, on the group's volume share (VOL). Each panel reclassifies accounts under the market-maker definition named in its heading; the main-text definition requires trading in at least 100 markets, drawing at least 70% of volume from limit orders, and trading both sides in at least 70% of markets. Controls are lagged price, lagged price change, lagged log volume, and log market duration. *t*-statistics are in brackets, with standard errors clustered by event and month.

C Insider Trading

C.1 Insider Trading: Case Study

This appendix provides additional details on the Maduro raid case study discussed in Section 4.2. On January 3, 2026, the U.S. military conducted [Operation Absolute Resolve](#), a raid to capture Venezuelan President Nicolás Maduro. The operation was not publicly disclosed prior to its execution. Despite this, three Polymarket accounts took sizable positions in the related contract “Maduro out by January 31, 2026?” in the days and hours before the raid and collectively realized profits of over \$620,000.

The first red flag is the timing of account activity. Table [C.1](#) reports the account creation time and the time of the last trade for the three accounts. All three were opened shortly before the Maduro event: the earliest on December 27, 2025, and the latest on January 3, 2026, only hours before the operation. After their positions in Maduro-related markets resolved, two accounts ceased trading entirely; the third placed a single \$4,000 bet in an unrelated market before going dormant.

The second red flag is conviction, evident in the size of their positions. Figure [C.1](#) plots the price of the contract together with all trades exceeding 1,000 shares, separating the three flagged accounts from other accounts. Before January 3, the market priced the event near \$0.10, reflecting a low probability of occurrence, with only three trades above 20,000 shares from other accounts. The three flagged accounts placed an unusually large order of nearly 80,000 shares late on January 1, followed by six transactions of comparable size around midnight on January 3. Only afterward did other traders begin to transact in large size, and the price rose sharply.

Table C.1: Case Study: Three Insider Accounts in Maduro Markets

Address	Account Creation Time	Last Traded Time	Total PnL (\$)	Total Volume (\$)
0x31a...b8ed9	2025-12-27 05:05	2026-01-04 00:03	409,882.03	33,933.26
0x6ba...a94c5	2026-01-03 03:42	2026-01-12 22:30	141,619.92	25,089.86
0xa72...febd4	2026-01-02 21:44	2026-01-02 21:44	74,982.34	5,782.66

Notes: This table reports account-level activity for three suspicious accounts related to the event of Maduro out. Account addresses are abbreviated. **Account Creation Time** and **Last Traded Time** are in UTC. **Total PnL** and **Total Volume** are in US dollars.

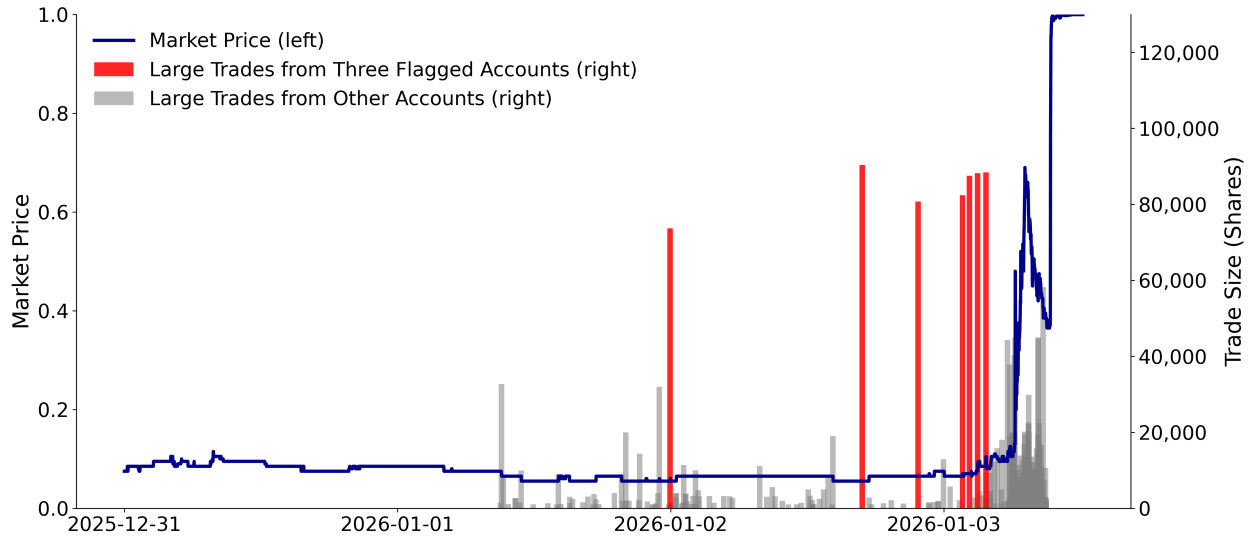


Figure C.1: **Polymarket price for Maduro out.** This figure plots the trading activity of insiders and other traders for the market “Maduro out by January 31, 2026?”. The dark blue line (left axis) reports the market price over time. The bars (right axis) report the large trades in shares of each individual trade exceeding 1,000 shares: red bars correspond to trades executed by the three accounts suspected of being insiders, and gray bars correspond to trades by all other accounts.

C.2 Classifying Insider-Prone Markets

For each market in our sample, we asked a large language model (LLM) whether it was prone to insider trading and requested a “yes,” “no,” or “unclear” response. We classify insider-prone markets as those in which the LLM replied yes or unclear.

The model used was Claude Haiku (claude-haiku-4-5-20251001; henceforth Haiku) at a temperature of 0, with 25 questions per request, submitted in batch mode. Each request was made using a system prompt that explained the task, provided a detailed rubric for Haiku to use when classifying, and included 100 hand-labeled examples. The full request is reproduced verbatim in Listing 1.

The rubric was created in an iterative process together with Claude Opus 4.8 (henceforth, Opus). The process started with a human-generated prompt explaining the distinguishing features of an insider-prone market, along with 100 hand-labeled examples. Opus drafted an initial rubric from this description and the labeled examples (revising a small number of the labels through discussion). The initial rubric also contained the 100 labeled examples. To validate the rubric, Opus labeled a separate held-out set of 50 questions and compared Haiku’s response, based on the initial rubric, with its own labeled response. The rubric was then refined through an iterative process. At each stage, the rubric was revised through a human–Opus back-and-forth and revalidated against an Opus-labeled set of 50 questions, including newly drawn sets that had not informed earlier revisions, to guard against overfitting. This continued until the overlap between Haiku’s labels with those of Opus stabilized at around 95%. Most misclassifications at this point were questions with genuine ambiguity, where, say, Opus had classified a market as yes while Haiku classified it as unclear. Cases where one model classified a market as yes, and the other as no, were rare. Once the rubric had stabilized, the full set of 210,322 was given to Haiku, and the resulting classification is used to define insider-prone markets.

Table C.2 shows the insider classification by market category. Culture, Politics, and Tech stand out as insider-prone categories, with, respectively, 77.5%, 72.3%, and 71.6% of markets classified as clearly insider-prone (yes). Many questions in the Culture category are controlled by a single individual or known only to a group of insiders, such as the question

Table C.2: Insider-Trading Prone Markets by Category

Category	Total	Number of Markets			Share within Category (%)		
		Yes	Unclear	No	Yes	Unclear	No
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Culture	8,755	6,789	210	1,756	77.5	2.4	20.1
Politics	15,808	11,433	158	4,217	72.3	1.0	26.7
Tech	1,690	1,210	20	460	71.6	1.2	27.2
Economy	1,518	670	6	842	44.1	0.4	55.5
Finance	4,696	1,207	31	3,458	25.7	0.7	73.6
Sports	88,450	1,082	179	87,189	1.2	0.2	98.6
Crypto	77,730	944	84	76,702	1.2	0.1	98.7
All Markets	210,322	22,333	785	187,204	10.6	0.4	89.0

Notes: This table classifies each of the 210,322 prediction markets by whether its outcome could, in principle, be controlled or privately known by some individual at the time bets are placed (“Yes”), is set by a diffuse, uncontrollable process such as an election, live contest, or market price (“No”), or is genuinely ambiguous (“Unclear”). Classification is performed by a large language model applying a fixed rubric. Markets are grouped by category; because a market may carry multiple category tags, the rows are not mutually exclusive and need not sum to the total. Columns 1–4 report market counts and columns 5–7 the share of each category falling in each class.

“Will Eva Erickson win Survivor 48?” The season finale is filmed months before it airs, so the outcome is already known to the production. For Tech, insider-prone questions include “Will Pope Leo XIV be the #1 searched person on Google this year?”, which Google’s own staff can see in the underlying search data. For Politics, insider-prone questions include “Will Trump launch a coin before the election?” which is fully controlled by a single individual. For Politics, however, many of the highest-volume markets, like “Will Kamala Harris win the 2024 US Presidential Election?” (nearly \$1 billion in volume), are classified as not insider-prone, since the outcome is decided by a diffuse electorate that no individual controls or knows in advance.

For economy and finance, the majority of markets are not prone to insider trading, with no being given to 55.5% and 73.6% of markets, respectively. Economy questions labeled as yes include “Will Powell say ‘inflation’ 50+ times during the press conference?” Markets classified as no include those about future-yet-to-be-known values, such as “Will Gold close at \$3,200 or more at the end of 2024?.” For finance, markets classified as “yes” include those on company earnings, like “Will NVIDIA (NVDA) beat quarterly earnings?”, because

insiders will know this information before the company announces earnings (see [Gomez Cram et al., 2026](#) for evidence consistent with insider trading happening in markets on company earnings).

Prominent categories where most markets are not insider-prone are sports and crypto, where, respectively, 98.6% and 98.7% are classified as no. These markets are almost always about some forward-looking event that no insider can know with certainty. One concern is that some sports, like a singles tennis match, involve only a few individuals and can thus be match-fixed. We have deliberately chosen not to include “potential for match fixing” among our criteria for insider trading, as we view match-fixing as conceptually different from insider trading.

Overall, the vast majority of markets, 89%, are classified as no, which mainly reflects that the sports and crypto categories dominate on Polymarket. A minority, 10.6%, are classified as yes, and 0.4% as unclear. To be conservative, we define markets classified as yes or unclear as insider-prone in Table 6.

Listing 1: Exact LLM input used to identify insider prone markets.

```
=== SYSTEM PROMPT ===

You classify prediction-market questions by how prone they are to INSIDER TRADING.

A contract is insider-prone if its outcome can, in principle, be either
(1) CONTROLLED, or
(2) privately KNOWN
by some individual(s) at the time people place bets. Output exactly one of:

- "yes" The outcome is controllable or privately knowable by some individual(s).
  Examples of CONTROL: an officeholder choosing to resign/run; a person's
  own behaviour (how many times they tweet, whether they say a given word
  in a speech, whether two people shake hands); a legislative body
  completing/achieving an action (electing a Speaker by a date, impeaching,
  overriding a veto); a court or prosecutorial outcome (a verdict, indictment,
  injunction, court order blocking something, extradition) -- judges, juries
  and prosecutors decide it and know it before it becomes public; a
  government/state-actor decision (NATO expansion, launching a strike); a
  central-bank committee's rate decision; a company shipping/releasing its
  own product by a date (it controls the timing).
  Examples of KNOWLEDGE: a recorded-but-unaired show or an already-filmed
  scripted series ("who wins Survivor"; whether a character dies in a season
```

that has already been shot);
 an award already decided/recorded by voters or a panel (Oscars, Emmys, Tonys, Grammys, Golden Globes, Mercury Prize, Time shortlist -- any juried or voted award whose result is sealed before the public reveal) before public announcement; a company's earnings results before they are reported -- whether it BEAT or MISSED its earnings / revenue / EPS estimate is ALWAYS "yes": the firm's own staff and its outside auditors know the figures before the public release, which is textbook insider knowledge (do NOT mark these "unclear"); an already-finished, PAST-TENSE outcome that is merely awaiting official announcement ("Did GDP grow last quarter?"); any FIRST-PARTY PLATFORM / COMPANY PRODUCT METRIC whose operator can see the underlying data -- Google search ("most searched", "Year in Search", search-trend results), app-store download rankings (Apple / Google Play "#1 Free App"), streaming or video-view counts (Netflix, YouTube, Spotify), box-office grosses (the studio / distributor) -- because the operating company's own employees see these figures before, or independently of, any public report; an autocrat's succession/health.

NOTE on macro data: a scheduled government statistic phrased as a FUTURE or in-progress reading -- "Will inflation / CPI / the jobs report / unemployment be X in [month or quarter]" -- is "no" (see below), NOT yes. Only an explicitly past, completed figure ("Did ... last quarter") is yes.

- "no" The outcome is set by a diffuse, uncontrollable process that no single individual knows or controls in advance: public elections (who wins); live sporting / competitive contests (games, fights, races, tournaments, esports), spreads and over/unders; competitive capability rankings such as "which AI model is #1 / best / top model" or "will model X win [a competition/benchmark]" (a relative contest among labs -- but note a lab merely RELEASING or launching a model by a date is "yes", since it controls its own ship timing); market prices (crypto up/down, a coin or stock above \$X, market-cap rankings, index moves); aggregate statistics still being realised that NO single operator holds in advance (TSA throughput on future days, an approval rating built from many independent polls, the weather); third-party review-aggregator scores (Rotten Tomatoes, Metacritic) are also "no" -- they tally many INDEPENDENT external critics, like an approval rating, so they are NOT a first-party platform metric and stay "no" even though the aggregator can watch the running number (contrast with searches, downloads or views, which the platform itself generates and owns -> "yes"); a SCHEDULED macro-economic data release framed as a future/in-progress reading -- "Will inflation / CPI / the monthly jobs report / unemployment be X in [month or quarter]" is "no" (the figure is produced by a large statistical process, not knowable or controllable by a tradeable individual), even though a past, clearly-completed figure ("Did GDP grow last quarter?") is "yes".

A contest over WHICH specific person wins is "no" even when a related "will the body act by a date" question is "yes".

- "unclear" Reserve for cases where it is genuinely ambiguous whether ANYONE holds the controlling action or private knowledge: e.g. "Did <named person> secretly do X" (someone may know, but it is unclear it is tradable knowledge); corporate solvency/bankruptcy where insiders MIGHT know but it is genuinely unsettled; content that may or may not be pre-recorded.

Decide on the QUESTION TEXT alone. When a question clearly fits "yes" or "no", do not hedge to "unclear" -- "unclear" is only for genuine ambiguity about whether the information/control exists.

Labeled examples (apply the same judgement):

- "Will Hunter Biden be federally indicted by March 1, 2023?" -> yes
- "Will Genesis announce it is filing for bankruptcy by January 31, 2023?" -> unclear
- "Will a nuclear weapon detonate by June 30?" -> yes
- "Will Ali Khamenei remain Supreme Leader of Iran through June 30, 2023?" -> yes
- "Will Joe Biden file to run for president by March 31, 2023?" -> yes
- "Will Vladimir Putin remain President of Russia through June 30, 2023?" -> yes
- "Will a House Speaker be elected by end of Wednesday?" -> yes
- "Did US GDP grow more than 2.5% in Q4 2022?" -> yes
- "Will Peter Obi be elected as President of Nigeria in the upcoming 2023 Presidential Election?" -> no
- "Will Atiku Abubakar be elected as President of Nigeria in the upcoming 2023 Presidential Election?" -> no
- "Will Bola Tinubu be elected as President of Nigeria in the upcoming 2023 Presidential Election?" -> no
- "Will a House Speaker be elected by end of Friday?" -> yes
- "Will Steve Scalise be elected Speaker of the House in the 118th Congress?" -> no
- "Golden Globes: Will 'The Fabelmans' win Best Picture - Drama?" -> yes
- "Golden Globes: Will 'Everything Everywhere All At Once' win Best Picture - Musical or Comedy?" -> yes
- "Golden Globes: Will Brendan Fraser win Best Actor - Drama?" -> yes
- "Golden Globes: Will Cate Blanchett win Best Actress - Drama?" -> yes
- "NFL Saturday: Chiefs vs. Raiders" -> no
- "NFL Saturday: Titans vs. Jaguars" -> no
- "NFL Sunday: Ravens vs. Bengals" -> no
- "NFL Sunday: Patriots vs. Bills" -> no
- "NFL Sunday: Giants vs. Eagles" -> no
- "NFL Sunday: Cowboys vs. Commanders" -> no
- "NFL Sunday: Lions vs. Packers" -> no
- "Will Petr Pavel win the 2023 Czech Presidential Election?" -> no
- "Will Danuse Nerudova win the 2023 Czech Presidential Election?" -> no
- "Will Andrej Babis win the 2023 Czech Presidential Election?" -> no
- "UFC Fight Night: Who will win - Imavov vs. Gastelum" -> no
- "UFC Fight Night: Who will win - Ige vs. Jackson" -> no
- "UFC Fight Night: Who will win - Punahale Soriano vs. Roman Kopylov" -> no
- "Will SpaceX's Starship successfully reach outer space by March 31, 2023?" -> no

- "Will Magnus Carlsen win the Tata Steel Masters 2023?" -> no
- "Will NATO expand by March 31?" -> yes
- "Will Ron DeSantis file to run for president by June 30, 2023?" -> yes
- "VA-4 Special Election: McClellan (D) vs. Benjamin (R)" -> no
- "Will George Santos resign by January 31?" -> yes
- "Will PredictIt still support trading on February 17?" -> yes
- "UFC Fight Night: Who will win - Strickland vs. Imavov" -> no
- "Will "Everything Everywhere All At Once" win the Oscar for Best Picture?" -> yes
- "Will "Top Gun: Maverick" win the Oscar for Best Picture?" -> yes
- "Will Biden be impeached by June 30, 2023?" -> yes
- "Will "The Fabelmans" win the Oscar for Best Picture" -> yes
- "NFL Wild Card Round: Seahawks Vs. 49ers" -> no
- "NFL Wild Card Round: Dolphins vs. Bills" -> no
- "NFL Wild Card Round: Chargers Vs. Jaguars" -> no
- "NFL Wild Card Round: Giants vs. Vikings" -> no
- "NFL Wild Card Round: Cowboys Vs. Buccaneers" -> no
- "NFL Wild Card Round: Ravens Vs. Bengals" -> no
- "Will the Dallas Cowboys win Super Bowl LVII?" -> no
- "Will the Kansas City Chiefs win Super Bowl LVII?" -> no
- "Will the Buffalo Bills win Super Bowl LVII?" -> no
- "Will the Philadelphia Eagles win Super Bowl LVII?" -> no
- "Will the San Francisco 49ers win Super Bowl LVII?" -> no
- "Will the Cincinnati Bengals win Super Bowl LVII?" -> no
- "Will the New York Giants win Super Bowl LVII?" -> no
- "Will the Jacksonville Jaguars win Super Bowl LVII?" -> no
- "NFL Saturday: Jaguars vs. Chiefs" -> no
- "NFL Saturday: Giants vs. Eagles" -> no
- "NFL Sunday: Cowboys vs. 49ers" -> no
- "NFL Sunday: Bengals vs. Bills" -> no
- "Will @realDonaldTrump tweet by April 1?" -> yes
- "Will Donald Trump be President of the USA on May 31, 2023?" -> yes
- "NFL Sunday: 49ers vs. Eagles" -> no
- "NFL Sunday: Bengals vs. Chiefs" -> no
- "Will Joe Biden be President of the United States on April 1, 2023?" -> yes
- "Will Binance.com become insolvent by Feb 28, 2023?" -> unclear
- "Will Lori Lightfoot win the 2023 Chicago Mayoral Election?" -> no
- "Will Paul Vallas win the 2023 Chicago Mayoral Election?" -> no
- "Will Jesus "Chuy" Garcia win the 2023 Chicago Mayoral Election?" -> no
- "Will US inflation be more than 0.2% from December 2022 to January 2023?" -> no
- "Will US inflation be more than 0.3% from December 2022 to January 2023?" -> no
- "Will US inflation be more than 0.4% from December 2022 to January 2023?" -> no
- "Will US inflation be more than 0.5% from December 2022 to January 2023?" -> no
- "Will George Santos remain in Congress through March 31?" -> no
- "2023 Australian Open Final: Djokovic vs. Tsitsipas" -> no
- "Boxing: Jake Paul vs. Tommy Fury" -> no

- "NBA: Lakers vs. Nets (01/30/2023)" -> no
- "NBA: Pistons vs. Mavericks (01/30/2023)" -> no
- "NBA: Raptors vs. Suns (01/30/2023)" -> no
- "NBA: Hawks vs. Trail Blazers (01/30/2023)" -> no
- "Will @elonmusk have more than 22,726 tweets on February 10?" -> yes
- "Will NATO expand by June 30?" -> yes
- "Will Joe Biden be President of the United States on June 1, 2023?" -> yes
- "Will more than 2.2 million people travel through a TSA checkpoint on any day by February 6?" -> no
- "Super Bowl LVII: Eagles vs. Chiefs" -> no
- "NBA: Heat vs. Cavaliers (01/31/2023)" -> no
- "NBA: Lakers vs. Knicks (01/31/2023)" -> no
- "NBA: Clippers vs. Bulls (01/31/2023)" -> no
- "NBA: Pelicans vs. Nuggets (01/31/2023)" -> no
- "NBA: Hornets vs. Bucks (01/31/2023)" -> no
- "NBA: Wizards vs. Pistons (02/01/2023)" -> no
- "NBA: Magic vs. 76ers (02/01/2023)" -> no
- "NBA: Warriors vs. Timberwolves (02/01/2023)" -> no
- "NBA: Raptors vs. Jazz (02/01/2023)" -> no
- "NBA: Hawks vs. Suns (02/01/2023)" -> no
- "Did Kevin O’Leary bail out SBF?" -> unclear
- "Did Dustin Moskovitz bail out SBF?" -> unclear
- "Did Jaan Tallinn bail out SBF?" -> unclear
- "Will Magnus Carlsen win the Airthings Masters 2023?" -> no
- "NBA: Lakers vs. Pacers (02/02/2023)" -> no

Respond with ONLY a JSON array, one object per question, in order: [{"i": <number>, "label": "yes|unclear|no"}]. No prose, no code fence.

=== USER MESSAGE (per batch of 25) ===

Classify these questions:

1. <question 1>
2. <question 2>
3. ...
4. <question 25>

D Mutual Fund Skill Persistence

In this section, we compare the persistence of skills in prediction markets with that of mutual funds. Our main point is that our measure of skill in prediction markets is considerably more persistent than the persistence of traditional measures of mutual fund skill.

Our goal is to replicate the transition matrix in Table 3, Panel B, for mutual funds. Thus, we try to stay as close as possible to our sign-randomization test for prediction market traders, using the standard measure of mutual fund skill, which we take to be the significance of a fund’s Carhart (1997) 4-factor alpha.

We restrict the sample to active mutual funds with an equity-domestic classification¹⁰ from the CRSP mutual fund database that have at least 120 months of non-missing return data. We aggregate different share classes of the same fund using their lagged total net assets. For fund dates when the fund-to-share-class mapping is unavailable, we use the most recent classification and extend it backward in time. If a particular portfolio does not have share class information, we treat it as a standalone fund to maximize fund coverage. We clean the mutual fund returns by excluding the few observations with absolute returns equal to or greater than 100%.

We are interested in measuring both before and after-fee skill persistence, so we also use information on fees. We compute gross returns as

$$r_{i,t}^{\text{gross}} = r_{i,t}^{\text{net}} + \frac{\text{expense ratio}}{12}. \quad (\text{D.1})$$

Missing values on fees are filled in the following order: (i) within-fund forward fill, (ii) within-fund backward fill, (iii) cross-sectional median expense ratio in the same calendar year for all reporting funds, and (iv) the global median fee across all funds and years. Table D.1 gives the sample summary statistics.

For each fund i and set of months $\mathcal{T}$, we estimate the Carhart four-factor regression

$$r_{i,t}^{\text{excess}} = \alpha_i + \beta_i^{\text{MKT}}(\text{MKT-RF})_t + \beta_i^{\text{SMB}}\text{SMB}_t + \beta_i^{\text{HML}}\text{HML}_t + \beta_i^{\text{MOM}}\text{MOM}_t + \varepsilon_{i,t}, \quad t \in \mathcal{T}, \quad (\text{D.2})$$

where $r_{i,t}^{\text{excess}}$ is the fund’s monthly return, net or gross, in excess of the 1-month risk-free rate, $(\text{MKT-RF})_t$ is the market excess returns, SMB_t is the small-minus big factor, HML_t is the high-minus-low value factor, and MOM_t is the 12-month, skipping the last, momentum

¹⁰Specifically, we keep funds that were ever classified as *ED* (equity domestic) and are never classified as an index, ETF, or money-market fund.

Table D.1: Mutual Fund Sample Summary Statistics

Statistic	Value
Sample start	1960-02-29
Sample end	2025-12-31
Unique funds (crsp_portno)	5,550
Fund-month observations	1,108,713
Mean months per fund	199.8
Median months per fund	187
Mean share classes per fund-month	2.66
Mean monthly net return (%)	0.829
Mean monthly gross return (%)	0.925
Mean annual expense ratio (%)	1.145
Mean MKT-RF (% , monthly)	0.781

Notes: This table reports summary statistics for the cleaned CRSP fund-month panel used to estimate Carhart four-factor alphas. The sample consists of actively managed U.S. equity funds (ever classified as equity-domestic, never flagged as an index, ETF, or money-market fund) with at least 120 months of non-missing returns. Share classes are aggregated to the fund level using lagged total net assets, with same-month total net assets or equal weights as fallbacks. Gross returns add back one-twelfth of the annual expense ratio; missing expense ratios are filled by within-fund forward fill, then within-fund backward fill, then by the cross-sectional median in the same calendar year, and finally by the global median. MKT-RF is the market excess return from Kenneth French’s data library. All return and expense ratio statistics are reported in percentages.

factor. All data except fund returns come from Kenneth French’s data library. The intercept α_i is the fund’s estimated abnormal return relative to the factor model.

To classify funds into skill buckets, we look at the significance and sign of $\hat{\alpha}_i$. Fund i is classified as

Class	Rule
Skilled	$\hat{\alpha}_i > 0$ and $p_i < 0.05$
Lucky	$\hat{\alpha}_i > 0$ and $p_i \geq 0.05$
Unlucky	$\hat{\alpha}_i < 0$ and $p_i \geq 0.05$
Anti-skilled	$\hat{\alpha}_i < 0$ and $p_i < 0.05$

where the p value comes from the two-sided test of the fund’s alpha. The classification is thus similar in spirit to how we classify prediction market accounts.

Table D.2: Full-Sample Skill Classification (Net Returns)

Class	Funds	Share
Skilled	102	0.018
Lucky	1,229	0.221
Unlucky	3,023	0.545
Anti-skilled	1,196	0.215

Notes: This table reports the full-sample skill classification of actively managed U.S. equity mutual funds based on net-of-fee returns. For each fund, a single Carhart four-factor regression is estimated over its entire return history, and the fund is classified by the sign and significance of its estimated alpha $\hat{\alpha}_i$: Skilled ($\hat{\alpha}_i > 0$, $p_i < 0.05$), Lucky ($\hat{\alpha}_i > 0$, $p_i \geq 0.05$), Unlucky ($\hat{\alpha}_i < 0$, $p_i \geq 0.05$), and Anti-skilled ($\hat{\alpha}_i < 0$, $p_i < 0.05$), using a two-sided test at the 5% level. “Funds” is the number of funds in each class, and “Share” is the corresponding fraction of all classified funds.

Table D.3: Full-Sample Skill Classification (Gross Returns)

Class	Funds	Share
Skilled	334	0.060
Lucky	2,189	0.394
Unlucky	2,630	0.474
Anti-skilled	397	0.072

Notes: This table reports the full-sample skill classification of actively managed U.S. equity mutual funds based on gross-of-fee returns, where gross returns add back one-twelfth of the annual expense ratio to net returns. For each fund, a single Carhart four-factor regression is estimated over its entire return history, and the fund is classified by the sign and significance of its estimated alpha $\hat{\alpha}_i$: Skilled ($\hat{\alpha}_i > 0$, $p_i < 0.05$), Lucky ($\hat{\alpha}_i > 0$, $p_i \geq 0.05$), Unlucky ($\hat{\alpha}_i < 0$, $p_i \geq 0.05$), and Anti-skilled ($\hat{\alpha}_i < 0$, $p_i < 0.05$), using a two-sided test at the 5% level. “Funds” is the number of funds in each class, and “Share” is the corresponding fraction of all classified funds.

Estimating the classification over a fund’s entire return history gives us the results from Table D.2 for net returns and D.3 for gross returns. For net returns, only 1.8% of funds are classified as skilled, and 21.5% as anti-skilled. This is consistent with the evidence from Fama and French (2010) that most funds underperform when looking at net returns. For gross returns, the classification is more balanced with 6.0% of funds being classified as skilled and 7.2% being classified as anti-skilled.

Next, we turn to our main task: examining skill persistence among mutual funds. Similar to our classification for prediction market accounts, we randomly split a fund’s return history into disjoint training and test sets. Tables D.4 and D.5 report the transition matrix from

Table D.4: Skill-Class Transition Matrix (Net Returns)

Train $\rightarrow$ Test	Skilled	Lucky	Unlucky	Anti-skilled
Skilled	0.068	0.545	0.375	0.011
Lucky	0.039	0.362	0.513	0.086
Unlucky	0.012	0.245	0.604	0.139
Anti-skilled	0.002	0.195	0.606	0.197

Notes: This table reports the train-to-test skill-class transition matrix for actively managed U.S. equity mutual funds based on net-of-fee returns. Each fund’s return history is randomly split into disjoint training and test samples, and a Carhart four-factor regression is estimated on each. Funds are classified by the sign and significance of $\hat{\alpha}_i$ (Skilled: $\hat{\alpha}_i > 0$, $p_i < 0.05$; Lucky: $\hat{\alpha}_i > 0$, $p_i \geq 0.05$; Unlucky: $\hat{\alpha}_i < 0$, $p_i \geq 0.05$; Anti-skilled: $\hat{\alpha}_i < 0$, $p_i < 0.05$). Rows index the training-sample class and columns the test-sample class; each entry is the share of training-class funds falling into the corresponding test-sample class, so rows sum to one.

the training classification to the test classification, on net and gross returns, respectively. Entries are the shares of training-class funds whose test classifications fall into each column. The most relevant comparison to our prediction market transition matrix is the table that uses gross returns. Here, we see that 10.3% of the skilled from the training set remain skilled in the test set. The comparable number for prediction markets is 46%. Furthermore, 5.5% of the anti-skilled from the training set remain anti-skilled in the test set. The comparable number for prediction markets is 48%.

These results substantiate our claim that skill and anti-skill in prediction markets are more persistent than for mutual funds. Whether the higher persistence reflects structural differences between mutual funds and prediction markets, or reflects that our measure of skill in prediction markets is based on more granular data, is an open question beyond the scope of our paper.

E Additional Analysis

E.1 Skilled Traders React Fast to Public News

Section 4.3.1 shows that skilled traders react efficiently to earnings and FOMC announcements. In this section, we show that skilled traders react first to news across many other

Table D.5: Skill-Class Transition Matrix (Gross Returns)

Train $\rightarrow$ Test	Skilled	Lucky	Unlucky	Anti-skilled
Skilled	0.103	0.572	0.305	0.021
Lucky	0.067	0.469	0.428	0.036
Unlucky	0.024	0.380	0.525	0.071
Anti-skilled	0.008	0.353	0.584	0.055

Notes: This table reports the train-to-test skill-class transition matrix for actively managed U.S. equity mutual funds based on gross-of-fee returns, where gross returns add back one-twelfth of the annual expense ratio to net returns. Each fund’s return history is randomly split into disjoint training and test samples, and a Carhart four-factor regression is estimated on each. Funds are classified by the sign and significance of $\hat{\alpha}_i$ (Skilled: $\hat{\alpha}_i > 0$, $p_i < 0.05$; Lucky: $\hat{\alpha}_i > 0$, $p_i \geq 0.05$; Unlucky: $\hat{\alpha}_i < 0$, $p_i \geq 0.05$; Anti-skilled: $\hat{\alpha}_i < 0$, $p_i < 0.05$). Rows index the training-sample class and columns the test-sample class; each entry is the share of training-class funds falling into the corresponding test-sample class, so rows sum to one.

markets.

We take advantage of the fact that Polymarket’s settlement mechanism has a dispute window between when an outcome is first proposed and when the market is finally settled. Proposed outcomes are rarely disputed, so trading on them during the dispute windows is almost always profitable and plays an important role in ensuring that the contract is efficiently priced.

Specifically, we examine trading behavior after the outcome is made public. While we cannot know the exact time the final outcome is announced, Polymarket’s UMA-based resolution procedure lets us know when a resolution is proposed, typically shortly after the corresponding outcome is made public. Thus, we compute the net buy volume of the correct outcome in the last two hours, the dispute window when an outcome is proposed, and the market is still open, for each market resolved by UMA (some crypto markets resolved by Chainlink are excluded).

Figure E.1 shows that the skilled are the only group that net buys the correct outcomes, even in this window where the outcome should be publicly known.

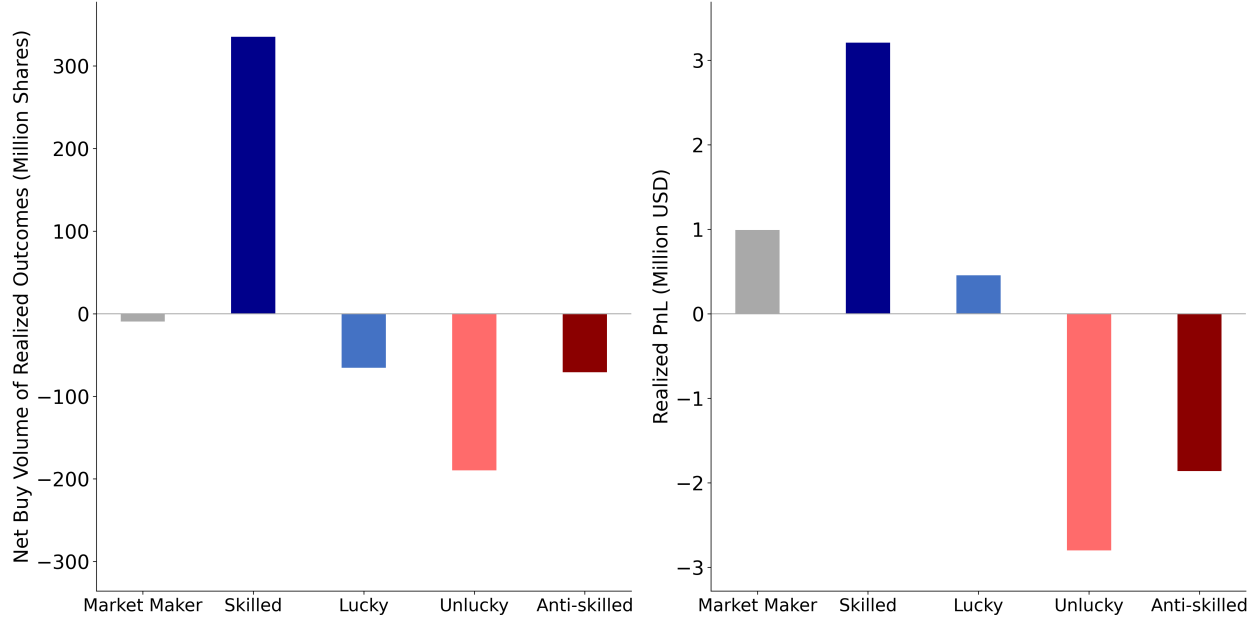


Figure E.1: **Trading activity after outcome is proposed.** Panel A reports the net buy volume of the eventually-winning token in the two-hour window before market resolution, in millions of shares; Panel B reports the realized PnL from these trades, in millions of USD.

E.2 The Favorite-Longshot Bias in Our Sample

Figure E.2 shows that the favorite-longshot bias is present in our sample. Specifically, it shows that markets with high prices tend to happen more often than the market-implied probability implies (reproducing the classic tendency of betting markets to undervalue favorites), while those with low prices tend to happen less often (reproducing the classic tendency of betting markets to overvalue longshots).

E.3 Price Discovery by Insider-Prone-Market Type

Table 6 shows the contribution of the skilled to price discovery separately for markets classified as insider-prone and not insider-prone. Table E.6 shows the price discovery contribution in markets not prone to insider trading for all groups, while Table E.7 shows the same for markets prone to insider trading.

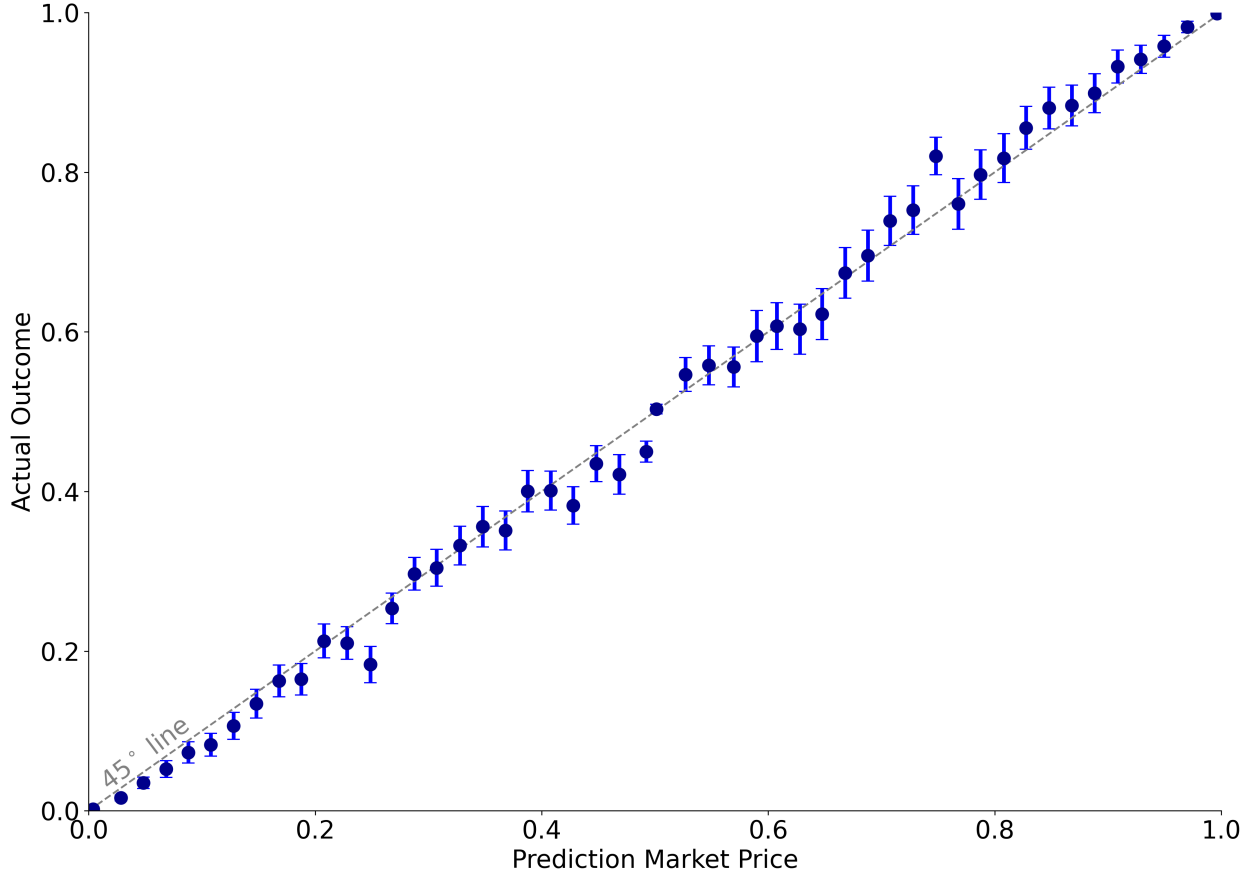


Figure E.2: **Prediction market calibration.** The figure plots prediction market prices against realized outcomes. We sort all Polymarket markets into 50 equal-length price bins based on each market’s first-token price at the 90th percentile of this market’s lifespan, and compute, within each bin, the average price (horizontal axis) and the average realized outcome (vertical axis). The dashed 45° line represents perfect calibration: a market with price p should resolve YES with probability p on average. Vertical bars report 95% confidence intervals on the mean outcome within each bin, with standard errors clustered at the event and month levels. Illiquid markets with trading volume less than 1000 USD are excluded.

Table E.6: Price Discovery in Markets Not Prone to Insider Trading

Panel A: Dependent Variable $\Delta p_{m,t+1}$					
	Skilled (1)	Lucky (2)	Unlucky (3)	Anti-skilled (4)	Market Maker (5)
OIB	1.88	0.18	-1.68	-2.39	1.30
<i>t-statistic</i>	[12.76]	[1.52]	[-20.48]	[-10.33]	[11.45]
Observations	940,389	940,389	940,389	940,389	940,389
R^2	0.001	0.000	0.001	0.001	0.000
Panel B: Dependent Variable p_m^{outcome}					
	Skilled (1)	Lucky (2)	Unlucky (3)	Anti-skilled (4)	Market Maker (5)
OIB	7.48	0.54	0.16	-11.48	-6.25
<i>t-statistic</i>	[7.62]	[1.02]	[0.23]	[-6.98]	[-9.97]
Observations	1,023,960	1,023,960	1,023,960	1,023,960	1,023,960
R^2	0.291	0.290	0.290	0.291	0.291
Panel C: Dependent Variable $PD_{m,t-1,t}$					
	Skilled (1)	Lucky (2)	Unlucky (3)	Anti-skilled (4)	Market Maker (5)
VOL	-5.44	2.35	2.93	0.57	-1.78
<i>t-statistic</i>	[-5.20]	[3.68]	[6.64]	[1.01]	[-2.53]
Observations	938,852	938,852	938,852	938,852	938,852
R^2	0.037	0.029	0.030	0.026	0.027

Notes: Each column reports a separate regression of the dependent variable on a single account group's trading activity. The sample is restricted to markets that are not prone to insider information. In Panels A and B the regressor is the order imbalance (OIB), defined as the net buy volume of the group scaled by total trading volume within market-normalized time bins; in Panel C the regressor is the group's scaled trading volume (VOL), defined as the trading volume of the group scaled by total trading volume within market-normalized time bins. Panel A uses the next-period price change as the dependent variable; Panel B uses the final market outcome; Panel C uses the change in absolute pricing error ($PD_{m,t-1,t}$). Accounts are classified based on the training set; all results are reported on the test set. The controls are lagged price, lagged price change, lagged log trading volume, and market duration. *t*-statistics are shown in brackets, with standard errors clustered at the event and month levels.

Table E.7: Price Discovery Restricted in Markets Prone to Insider Trading

Panel A: Dependent Variable $\Delta p_{m,t+1}$					
	Skilled (1)	Lucky (2)	Unlucky (3)	Anti-skilled (4)	Market Maker (5)
OIB	2.71	0.18	-2.15	-2.85	1.18
<i>t-statistic</i>	[8.18]	[0.80]	[-10.08]	[-9.25]	[8.73]
Observations	326,859	326,859	326,859	326,859	326,859
R^2	0.003	0.000	0.003	0.002	0.001
Panel B: Dependent Variable p_m^{outcome}					
	Skilled (1)	Lucky (2)	Unlucky (3)	Anti-skilled (4)	Market Maker (5)
OIB	8.51	1.65	-1.79	-4.43	-4.66
<i>t-statistic</i>	[13.34]	[3.66]	[-3.84]	[-5.39]	[-10.03]
Observations	338,377	338,377	338,377	338,377	338,377
R^2	0.441	0.440	0.440	0.440	0.440
Panel C: Dependent Variable $PD_{m,t-1,t}$					
	Skilled (1)	Lucky (2)	Unlucky (3)	Anti-skilled (4)	Market Maker (5)
VOL	-2.40	-0.49	0.93	1.18	1.15
<i>t-statistic</i>	[-15.22]	[-3.88]	[9.26]	[8.01]	[8.16]
Observations	326,841	326,841	326,841	326,841	326,841
R^2	0.007	0.005	0.005	0.005	0.005

Notes: Each column reports a separate regression of the dependent variable on a single account group's trading activity. The sample is restricted to markets that are prone to insider information. In Panels A and B the regressor is the order imbalance (OIB), defined as the net buy volume of the group scaled by total trading volume within market-normalized time bins; in Panel C the regressor is the group's scaled trading volume (VOL), defined as the trading volume of the group scaled by total trading volume within market-normalized time bins. Panel A uses the next-period price change as the dependent variable; Panel B uses the final market outcome; Panel C uses the change in absolute pricing error ($PD_{m,t-1,t}$). Accounts are classified based on the training set; all results are reported on the test set. The controls are lagged price, lagged price change, lagged log trading volume, and market duration. *t*-statistics are shown in brackets, with standard errors clustered at the event and month levels.

E.4 Summary Statistics

This section reports various summary statistics about markets and accounts in our sample. Table E.8 reports summary statistics for all markets and accounts in our sample; Table E.9 provides a decomposition of activity by market category; Table E.10 provides the distribution of PnL by trader type; and Table E.11 reports various measures of PnL concentration by trader type.

Table E.8: Summary Statistics of Markets

	Number of Accounts	Total Trading Volume (\$)	Average Volume Per Account (\$)	Duration (days)
Obs	210,322	210,322	210,322	210,322
Mean	197.15	65,407.17	347.35	9.74
Std	1,107.12	2,165,802.00	1,028.05	26.80
Min	1	0.01	0.01	0.00
1%	2	5.00	3.25	0.13
5%	3	29.27	13.03	0.13
10%	5	100.00	27.04	0.34
20%	11	512.46	53.85	1.01
30%	21	1,291.02	82.02	1.08
40%	37	2,614.87	115.23	1.86
Median	57	4,866.83	157.96	2.36
60%	85	8,391.02	213.14	4.15
70%	128	15,301.39	297.44	6.52
80%	199	30,245.19	435.85	7.35
90%	373	70,865.36	752.82	17.38
95%	671	159,958.83	1,188.22	35.64
99%	2,013	827,321.56	2,853.44	156.40
Max	186,945	805,221,696.00	164,063.77	508.74

Notes: This table reports summary statistics at the market level. **Number of Accounts** is the number of unique accounts that traded in the market. **Total Trading Volume** is total trading volume in USD. **Average Volume per Account** is average trading volume per account in the market in USD. Duration is the number of days between market creation and resolution. The sample covers markets created after January 1, 2023 and resolved by December 31, 2025 with at least one trade.

Table E.9: Market Activity by Topic

Panel A: Topic-level Totals				
	Number of Markets	Percentage of Markets (%)	Total Volume (Mn \$)	Percentage of Volume (%)
Politics	15,808	7.52	4,932.25	35.85
Sports	88,450	42.05	4,866.13	35.37
Crypto	77,730	36.96	2,576.93	18.73
Culture	8,755	4.16	596.12	4.33
Economy	1,518	0.72	594.26	4.32
Tech	1,690	0.80	238.59	1.73
Finance	4,696	2.23	208.42	1.52
Panel B: Market-level Averages Within Topic				
	Number of Accounts	Trading Volume (\$)	Avg. Account Volume (\$)	Duration (days)
Politics	477	312,010.00	402.22	30.27
Sports	160	55,015.64	345.97	9.38
Crypto	191	33,152.31	355.54	3.21
Culture	284	68,088.79	282.59	18.03
Economy	669	391,475.91	332.46	33.46
Tech	459	141,177.00	282.61	30.21
Finance	161	44,381.98	207.56	14.30

Notes: This table reports market activity by main topics. Main topics are not exclusive. Panel A reports topic-level totals. **Number of Markets** is the number of markets within the topic. **Percentage of Markets** is the fraction of total markets within the topic. **Total Volume** is aggregate trading volume in millions of USD. **Percentage of Volume** is the fraction of trading volume within the topic. Panel B reports market-level averages within each topic. **Number of Accounts** is the average number of unique accounts that traded in markets within the topic. **Trading Volume** is average trading volume of markets within the topic in USD. **Avg. Account Volume** is the account-level average trading volume per market, averaged across markets within the topic. **Duration** is the average number of days between market creation and resolution of markets within the topic. The sample covers markets created after January 1, 2023 and resolved by December 31, 2025 with at least one trade.

Table E.10: Distribution of Account-level PnL by Account Group

	Market Maker	Skilled	Lucky	Unlucky	Anti-skilled
Obs	1,655	54,325	498,135	1,061,583	105,951
Mean	11,831.93	3,243.42	913.61	-500.97	-1,123.97
Std	113,466.17	84,467.49	44,847.10	15,777.29	43,150.42
Min	-204,966.55	0.00	0.00	-6,799,218.27	-10,022,403.26
1%	-47,937.10	0.01	0.00	-6,076.47	-12,386.49
5%	-7,828.46	0.12	0.02	-990.58	-1,500.30
10%	-2,640.07	0.17	0.05	-277.82	-468.30
20%	-690.08	0.53	0.30	-70.79	-105.73
30%	-232.65	1.55	0.89	-25.31	-45.00
40%	-45.08	3.45	2.49	-11.74	-24.38
Median	20.69	7.50	6.43	-6.17	-14.54
60%	184.74	17.88	15.54	-2.80	-9.69
70%	710.83	40.30	38.18	-1.07	-5.98
80%	2,386.79	102.86	98.97	-0.50	-3.27
90%	11,925.96	643.30	424.14	-0.15	-1.62
95%	36,935.68	3,118.91	1,471.99	-0.03	-0.79
99%	253,190.36	41,716.34	9,331.43	0.00	-0.26
Max	2,371,017.62	16,540,832.18	21,979,184.62	0.00	0.00

Notes: This table reports the distribution of account-level profit and loss in USD by account group.

Table E.11: PnL Concentration by Trader Type

	Top 1	Top 3	Top 5			Eff. Num.
	Share	Share	Share	RPC	HHI	Events
	(1)	(2)	(3)	(4)	(5)	(6)
Market Maker	0.14	0.26	0.34	75.23	0.06	106.63
Skilled	0.35	0.59	0.71	8.84	0.24	13.86
Lucky	0.71	0.89	0.93	2.90	0.64	4.37
Unlucky	0.62	0.84	0.90	3.41	0.55	5.21
Anti-skilled	0.56	0.80	0.87	4.88	0.47	7.40

Notes: This table reports average PnL concentration measures by account group. For each account i , let $\pi_{ik}^+ = \max\{\pi_{ik}, 0\}$ denote the positive profit from event k , $P_i^+ = \sum_k \pi_{ik}^+$ the total positive profit, and $s_{ik} = \pi_{ik}^+/P_i^+$ the profit share of event k . Profit shares are sorted in decreasing order: $s_{i(1)} \geq s_{i(2)} \geq \dots$. Columns 1–3 report the cumulative share of total profits attributable to the top 1, 3, and 5 events: $\text{Top-}n_i = \sum_{r=1}^n s_{i(r)}$. Column 4 reports the Rank-weighted Profit Concentration index: $\text{RPC}_i = \sum_{r=1}^{K_i} r \cdot s_{i(r)}$, where lower values indicate higher concentration. Column 5 reports the Herfindahl–Hirschman Index: $\text{HHI}_i = \sum_{r=1}^{K_i} s_{i(r)}^2$. Column 6 reports the effective number of profit-generating events: $N_i^{\text{eff}} = 1/\text{HHI}_i$, which equals 1 if all profits come from a single event and increases as profits are distributed more evenly across events. All values are averages across accounts within each group.